EcoFog: A Novel Composite Mesh for Passive Water Collection from Fog and High-Relative Humidity Environments

Mrityunjay Gupta, Diivij Todi

Abstract

Freshwater scarcity remains a critical global public health challenge, with over 2 billion people lacking access to safe drinking water. In arid and semi-arid regions, fog harvesting has emerged as a promising solution in which atmospheric water droplets are passively condensed and collected on mesh surfaces. However, the only widely implemented system, the Raschel mesh, currently fails to maximise water collection efficiency, suffering from persistent clogging. In recent years, alternative systems have been explored, such as the Janus and biomimetic meshes, that outperform the Raschel mesh under simulated fog conditions, but rely on wind regime and complex fabrication processes, ultimately rendering large-scale employment of such systems unfeasible. These limitations present the need for a passive fog collector that sustains high water-collection efficiency over prolonged periods while being scalable. Hence, we developed EcoFog, an environmentally-friendly coplanar composite mesh combining a hydrophobic backbone with electrospun hydrophilic nanofibres and nanoparticles to simultaneously achieve high nucleation density and rapid drainage of water droplets.

EcoFog was fabricated by 3D-printing a Polylactic Acid (PLA) mesh, PLA being selected for its high hydrophobicity, with a shade coefficient (the fraction of total area occupied by the material) optimised at 38.8%. Then, a hydrophilic matrix of Polyvinyl Alcohol (PVA) and Polyethene Oxide (PEO) nanofibers was electrospun between the mesh's air gaps, with PVA providing nucleation sites and PEO improving solution rheology and jet stability. The matrix was then functionalized with Titanium Dioxide (TiO_2) nanoparticles at an optimised loading of 1 wt%, enhancing hydrophilicity by lowering the free energy barrier for nucleation. To increase matrix durability, Colloidal Silica (SiO_2) nanoparticles at an optimised loading of 1 wt% were introduced, distributing the mechanical stress on the matrix.

During field testing conducted at ~90-100% relative humidity, ~10-25°C, and wind speeds of 2-5 m/s, EcoFog achieved a mean water collection rate of 17.3 ± 2.1 $L/m^2/day$, corresponding to a 3.46-fold improvement over the Raschel mesh standard. The system's nucleation density was 3.8 times higher than that of the Raschel mesh, while the retained-water fraction (the fraction of total collected water that was retained in the mesh) decreased by 50%, indicating reduced clogging. Furthermore, the materials characterisation confirmed our intended surface chemistry: FESEM imaging revealed continuous,

bead-free nanofiber networks; sessile-drop goniometry (contact-angle testing) verified the wettability contrast; FTIR confirmed acetal bridge formation; and XRD analysis showed an amorphous structure.

Alongside experimental optimisation, we explored Gaussian Process regression using 36 preliminary measurements to investigate the potential of machine learning in predicting material performance and guiding experimental selection. This exploratory framework highlighted opportunities for uncertainty-aware Bayesian optimisation, offering a computational direction for reducing experimental demands across larger design spaces.

By optimising the surface chemistry and creating a unique wettability contrast within the coplanar amphiphilic architecture, EcoFog enables a cycle of continuous droplet nucleation, coalescence, and rapid drainage. This allows the mesh to improve collection efficiency without relying on wind-dependent geometries or complex fabrication, thereby bridging the gap between high-efficiency fog harvesting and practical deployment in arid, resource-constrained communities worldwide.

Keywords*: fog harvesting, wettability contrast, electrospinning, nanofiber composites, polylactic acid, polyvinyl alcohol, titanium dioxide, coplanar architecture*

Acknowledgment

We would like to gratefully acknowledge the support of several institutions that have made the described experiments possible. The Robotics Laboratory at Jayshree Periwal International School provided access to its 3D-printing facilities, which were essential for the fabrication of the PLA support structures. Similarly, the school's Chemistry Laboratory provided the facilities necessary to prepare the polymer solutions, to treat the spun solutions with NaCl and glutaraldehyde, and to perform the necessary washing steps. We would also like to thank the team at Patwa Hospital for their supervision of the electrospinning work. We would like to thank the team at Manipal Academy of Higher Education, Manipal, for providing access to the materials characterisation facilities in their laboratories, as well as for the guidance provided to us. We would like to thank the organising committee of the IRIS National Fair for allowing us to present this project. We are grateful for the advice and feedback received from Dr Kaushik Pal, IIT Roorkee and Dr S. Ranganathan. Furthermore, the project was presented at the Regeneron International Science and Engineering Fair 2026, where it received the Third Grand Award in the category of Materials Science. The present research has undergone additional experiments and analysis following these presentations.

Table of Contents

Abstract **2**
Acknowledgment **3**
1. Introduction **7**
1.1 Motivation 7
1.2 Literature Review 7
1.3 Aim and Research Hypothesis 9
1.4 Novelty 9
1.5 Material Selection 10
1.5.1 UV-stabilised Poly(lactic acid) PLA Structural Phase 10
1.5.2 PVA Hydrophilic Capture Phase 11
1.5.3 PEO Rheology Modifier 11
1.5.4 Rutile TiO_2 Surface-active Phase 11
1.5.5 Amorphous SiO_2 Secondary Phase 12
1.5.6 Glutaraldehyde Crosslinking Chemistry 12
2. Methodology **13**
2.1 PLA Backbone 13
2.2 Electrospinning 15
2.3 TiO2 Concentration Optimization 17
2.4 SiO2 Concentration Optimization 18
2.5 Hydrated Tensile Strength Testing 18
2.6 Combined Nanofiller Concentration Optimization 19
2.7 Glutaraldehyde Crosslinking 20
2.8 Material Characterization 21
2.8.1 Field-emission Scanning Electron Microscopy (FESEM) 21
2.8.2 Energy Dispersive X-ray Spectroscopy 21
2.8.3 Fourier-transform Infrared Spectroscopy 22
2.8.4 X-ray Diffraction 22
2.8.5 Sessile Drop Contact Angle 22
2.9 Fog Water Collection 22
2.9.1 Fog chamber Design 22
2.9.2 Outdoor Testing 24
3. Results **25**
3.1 Material Characterization 25
3.1.1 Field Emission Scanning Electron Microscopy (FESEM) 25
3.1.2 Contact Angle 27
3.1.3 FTIR Analysis of Polymer Network 29
3.1.4 Energy-dispersive X-ray Spectroscopy (EDS) 30
3.1.5 X-ray Diffraction (XRD) 31
3.2 Multiscale Optimisation of Collector Architecture 32

3.2.1 Support Geometry and Shade Coefficient 32
3.2.2 Polymer Concentration and PVA/PEO Ratio 33
3.2.3 Spinning Distance 34
3.2.4 TiO2 Loading 35
3.2.5 SiO2 Loading 36
3.2.6 Total Oxide Loading and Final Formulation 36
3.3 Controlled Fog-chamber Performance 37
3.4 Outdoor Validation and Water Recovery 38
3.4.1 Field Performance 38
3.5 Limitations 40
4. Conclusion **40**
5. References **42**
6. Appendix **46**
7. Lab Logbook **52**

1.Introduction

1.1 Motivation

According to the United Nations International Children's Emergency Fund (UNICEF, 2025), over 2 billion people worldwide lack access to fresh water [1]. The majority of the population lacking access to safe drinking water lives in rural, remote, and geographically vulnerable areas [1]. This disparity thus demands the development of alternative water sources, such as groundwater, rainwater, and fog harvesting [2].

In coastal mountains and highland areas that experience fog, water could be collected through a technique called fog harvesting [2]. In this method of collecting atmospheric water, a porous collecting structure is placed directly within the fog. The moisture in the fog droplets lands on the structure, and the collected water then drains off [3]. Thus, fog and atmospheric harvesting present a promising means of procuring water with minimal energy input [2]. Fog collectors have already been implemented in communities across Chile, Morocco and other regions that experience fog [2,5], with the most prominent example being the CloudFisher installation in southwest Morocco [5].

1.2 Literature Review

The standard system for fog collection was formalised by Schemenauer and Cereceda in 1994 [3]. These are Raschel-type polypropylene or High-Density Polyethene (HDPE) meshes commonly used due to their low cost, availability and simplicity of design [3,4]. However, the limitation of these materials relates not to their manufacturability but to their physical and chemical characteristics in relation to the task of fog collection, with efficiencies as low as 3-10 L m^{-2} day^{-1} [2,4].

In recent years, however, high-performance collecting materials have been explored in research. For instance, nanofibre coatings increase the density of small collecting sites, contributing towards a higher specific area for droplet impaction [9], while the Janus architectures incorporate materials with particular

wettability characteristics [10,14]. Other designs utilise complex three-dimensional kirigami structures to direct the fog flow towards the mesh [11]. Finally, electrostatic collectors leverage static charges to enable fog droplets to be collected by the harvester [12].

While the contemporary literature showcases potential for advancement, they also explain the trade-offs that often accompany these modifications. The high capture efficiency of nanofibre-coated surfaces introduces an aerodynamic penalty: although the addition of nanofibres increases the interception area [9], an excessively high specific surface area has been documented to increase the solid fraction, leading to the formation of water bridges over pores within the nanofibre layer [39]. This introduces a pressure drop along the airflow and diverts the airstream around the water collector [6,39].

These effects were also demonstrated by Lee et al. for the Janus concept, where one side is intentionally hydrophilic, and the other is hydrophobic. In such meshes, the same parameters that enhance the collection of water from fog can also hinder such collection [10], as the high surface area of the nanofibres increases the possibility of water retention [9.39]. Additionally, Janus meshes are flux-dependent, since they require the directional flow of wind from the hydrophilic side to the hydrophobic side. Hence, this design has been documented to lose efficiency in real-world environments [10].

Other collectors that exhibit high water collection efficiency also require additional infrastructure or treatments to the nanofibres. For example, electrostatic collection of water droplets requires an application of a high-voltage field, which prevents deployment in countries lacking access to a continuous power supply [12], and the use of a fluorinated low-energy coating [11] fails to uphold the environmental feasibility of the mesh [22].

Additionally, biomimetic techniques have been explored, taking inspiration from biological systems [4]. Yet, they are difficult to scale since they require fabrication of complex structures like microfabrics and three-dimensional nanostructures [4, 26].

To therefore develop an efficient and scalable fog harvester, four constraints must be addressed simultaneously: high droplet interception, efficient water recovery, a credible manufacturing route, and choices of materials that do not undermine its environmental and economic impact.

1.3 Aim and Research Hypothesis

We hypothesise that by developing a coplanar Janus architecture using commonly used materials and techniques, a scalable fog-harvesting surface could be developed that achieves a higher water collection efficiency than current standards. Utilising a PVA/PEO matrix could increase the rate of water droplet capture as compared to the Raschel mesh, and incorporating a PLA backbone could create a wettability gradient from hydrophilic to hydrophobic, hence preventing pore clogging.

The hypothesis is tested through the optimization of the PLA lattice structure, nanofibre composition, and oxide functionality, and through the assessment of the nanofibres through microscopy, spectroscopy, diffraction analyses, wettability analyses, fog droplet capture tests, and outdoor tests alongside PLA and Raschel mesh controls of identical geometry under identical conditions.

1.4 Novelty

EcoFog introduces a multiscale-optimised coplanar hetero-wetting architecture. The hydrophilic electrospun PVA/PEO nanofibres, coupled with hydrophilic TiO_2 nanoparticles, provide sites for sub-micrometre fog droplets to be captured by the collector [9,15], while the less-wettable PLA lattice allows for drainage of condensed water without requiring a front-and-back Janus structure [10]. Each element of the coplanar collector was jointly optimised to provide an efficient design that balances each of these components at multiple length scales. This ultimately enables the capture of microscopic droplets, rapid coalescence, and finally, instantaneous drainage

Furthermore, EcoFog follows a deployment-centric approach. The use of aqueous electrospinning to create the nanofibres leads to lower chemical oxide loadings, while completely avoiding fluorine-based

chemistry [22] and the requirement for no external energy to operate the collectors [12]. Additionally, the production of these collectors can be scaled industrially to ensure cost-effectiveness [23]. Overall, the integration of these various aspects into a single collector design distinguishes EcoFog from other fog collectors as both scalable and efficient, thereby enabling large-scale adoption.

Beyond its material architecture, EcoFog also provided an opportunity to investigate the intersection of computational optimization and experimental materials science. A preliminary Gaussian Process framework was explored to assess how machine learning could capture relationships between fabrication parameters and water collection, potentially informing more efficient experimental design in future iterations [47].

1.5 Material Selection

1.5.1 UV-stabilised Poly(lactic acid) PLA Structural Phase

Poly(lactic acid) (PLA) was chosen as the structural component due to its high level of rigidity, hydrophobicity, thermoplasticity, and lower affinity for water in comparison to the PVA-rich nanofibre phase [28]. PLA contains ester groups and non-polar methyl ($–CH_3$) groups but lacks the dense hydroxyl functionality of PVA [32], limiting extensive hydrogen bonding with water. This lower surface affinity promotes droplet beading and gravity-assisted runoff rather than prolonged water retention [10], allowing the PLA lattice to provide the hydrophobic drainage pathway while the PVA-rich fibres perform the principal capture function.

PLA was also selected because it is one of the most widely used bio-based plastics; it is commonly available both for additive manufacturing and large-scale processing [29]. Alternatives to PLA include biodegradable PCL plastics, which have lower stiffness [30]; commonly used PETG and ABS plastics, which are petroleum-based but not readily biodegradable [24]; and PP, which is more hydrophobic but more difficult to process [31].

1.5.2 PVA Hydrophilic Capture Phase

Poly(vinyl alcohol) (PVA) was selected as the primary capture polymer due to its dense (–OH) hydroxyl groups along its backbone [32]. These act as hydrogen-bond donors and acceptors, interacting strongly with water molecules, leading to high hydrophilicity [32,18]. This hydrophilicity contrasts with the relatively hydrophobic PLA lattice [28].

1.5.3 PEO Rheology Modifier

Poly(ethylene oxide) (PEO) was included in the system in small amounts to improve fibre formation [34]. The ether groups (–CH_2–CH_2–O–) within the PEO molecules can form hydrogen bonds with the hydroxyl groups of the PVA, leading to good compatibility between the two polymers [33]. Additionally, the high molecular weight of PEO increases the chain entanglement and extensional viscosity of the system, which helps maintain a continuous polymer jet and reduce the formation of beads [35]. PEO therefore functions mainly as a rheological modifier.

1.5.4 Rutile TiO_2 Surface-active Phase

Rutile TiO_2 of nominally 20 nm was chosen at low loading as the main inorganic component responsible for the surfactant properties of the system. The Ti-O groups can interact with water to form Ti-OH groups [27,36], permitting water-interaction sites for smaller water droplets. The introduction of TiO_2 particles increases the number of these groups in the system [36]. Rutile is preferred to anatase since it has lower photocatalytic activity under UV light, which would otherwise lead to degradation of the polymer components [20]. The rutile phase remains industrially abundant and chemically stable [36]. Furthermore, the choice of ZnO as the principal oxide was rejected due to its significant sensitivity to pH in its dissolution to release zinc ions into aqueous environments [21]. Al_2O_3 is another oxide that is stable in aqueous environments [37]. However, TiO_2 is utilised in a variety of polymers and is available from a variety of industrial sources [20]. Thus, a loading study based upon TiO_2 concentration was performed.

1.5.5 Amorphous SiO_2 Secondary Phase

Colloidal SiO_2 of approximately 30 nm diameter was incorporated as a secondary phase of low loading [19]. Since the silanol groups (Si-OH) present on the amorphous SiO_2 can form hydrogen bonds with the hydroxyl groups of the PVA [19], the polymer interacts with the particles [38]. These interactions can contribute to the structural integrity of the resulting fibre network. Thus, SiO_2 acts as a network modifier for the fibre.

1.5.6 Glutaraldehyde Crosslinking Chemistry

Glutaraldehyde was chosen as the crosslinker for the system due to its ability to effectively stabilise the water-soluble PVA polymer. Under acid-catalysed conditions, the aldehyde groups of glutaraldehyde react with the hydroxyl groups of PVA to form covalent linkages between PVA chains [16-18]. Such crosslinking reduces the mobility of the PVA chains and reduces their potential to dissolve in the surrounding water.

Only a proportion of hydroxyl groups in the PVA are crosslinked, allowing for the remainder of the PVA chains to remain hydrophilic. Thus, glutaraldehyde balances the need for structural stability with the requirement for the PVA to remain hydrophilic.

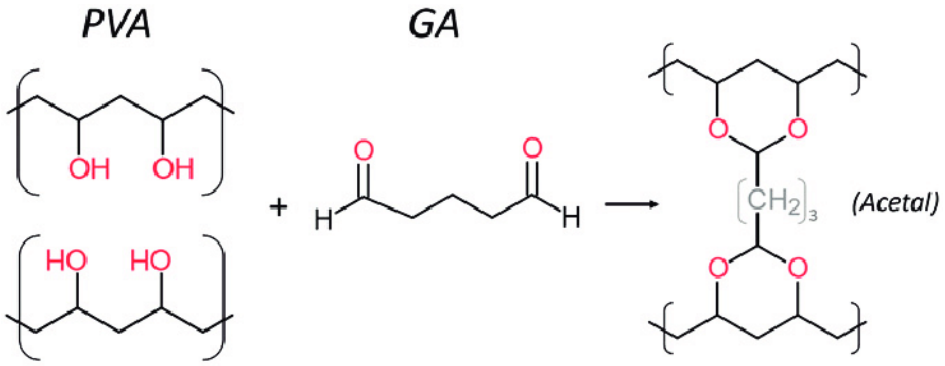


Figure 1. Reaction of crosslinking between PVA and glutaraldehyde [40]

2. Methodology

2.1 PLA Backbone

The optimal shape and dimensions for the PLA backbone were investigated. Multiple designs were tested by measuring their water collection in order to determine an optimum shape for the backbone.

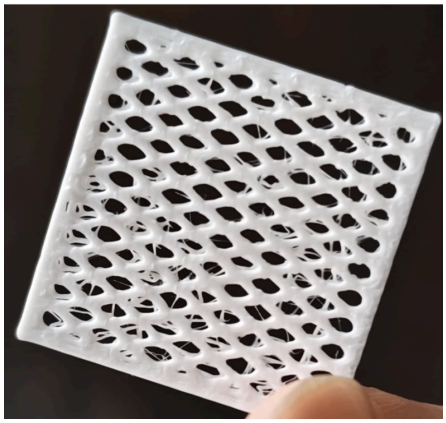


Figure 2. Initial PLA support design 1

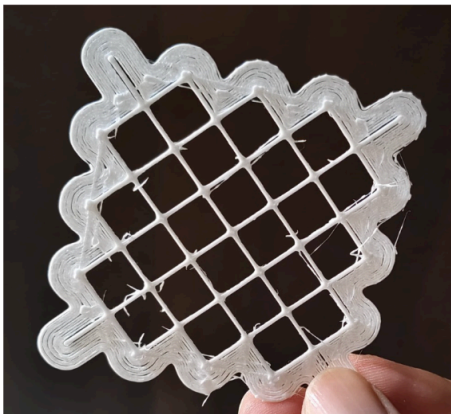


Figure 3. Initial PLA support design 2

The optimum-shaped backbone was designed in CAS and resembled a repeating trapezium unit cell mesh. The dimensions of the backbone were optimised to allow aerodynamic access and interception area. The shade coefficient, defined as the projected solid fraction of the mesh, of the optimised backbone was calculated using:

$$SC = s = \frac{A_{Blocked}}{A_{Total}}$$

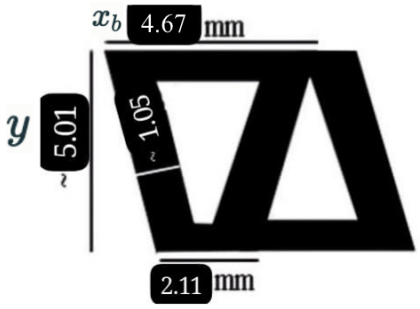


Figure 4. Optimised design and dimensions for PLA support

Using the formula for shade coefficient for a trapezium unit cell geometry [41], the shade coefficient was calculated using:

$$s = 1 - \frac{y(x_s + x_b)(b - t)}{2b(y + t)(t + \frac{x_s + x_b}{2})}$$

Where x_b is trapezium base width, x_s is trapezium top width, y is vertical spacing, t is strand thickness, and b is total mesh width. This computation was used as a quantitative design constraint prior to fabrication to ensure repeatable geometry.

The backbone was then fabricated using fused deposition modeling using PLA. Post-printing, each mesh was inspected for strand continuity, absence of missing segments, and uniformity of pores. Defects were identified and discarded as they interfere with aerodynamic capture.

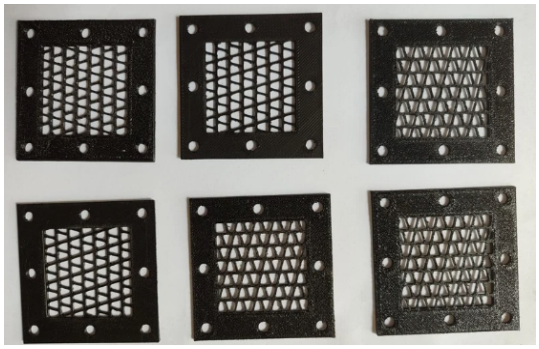


Figure 5. Finalised PLA backbones

2.2 Electrospinning

The electrospinning dope was formulated to produce the desired hydrophilic nanofiber network in order to increase nucleation site density. PVA was dissolved in 100 cm^3 of distilled water to swell the PVA. The solution was then heated to 90℃ under stirring. Then the solution was cooled to 60℃, and the PEO was dissolved into the solution. The nanoparticles TiO_2 and SiO_2 were introduced as a dispersion. The mixture was then kept under stirring and then ultrasonicated.

Figure 6. Electrospinning dope preparation using an oil bath, stirring at 490 rpm.

The backbones were then pre-treated with 1 M NaCl for 15 minutes, and then oven-dried for 15 minutes. This was done to increase the surface conductivity and improve the coating conformity.

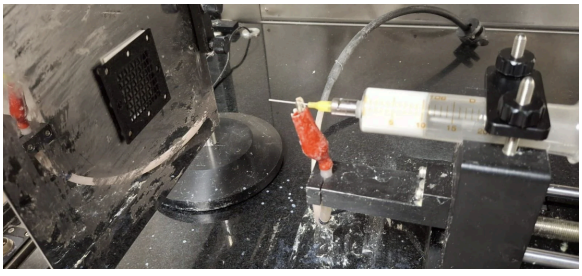


Figure 7. Electrospinning apparatus used for nanofibre deposition onto the PLA backbone.

Single-needle electrospinning was utilised with a static collector setup with an 18G needle. The applied voltage was set to 18 to 20 kV. The tip-to-collector distance was optimised at a distance of 7.5 cm. The deposition time was set to 30 minutes.

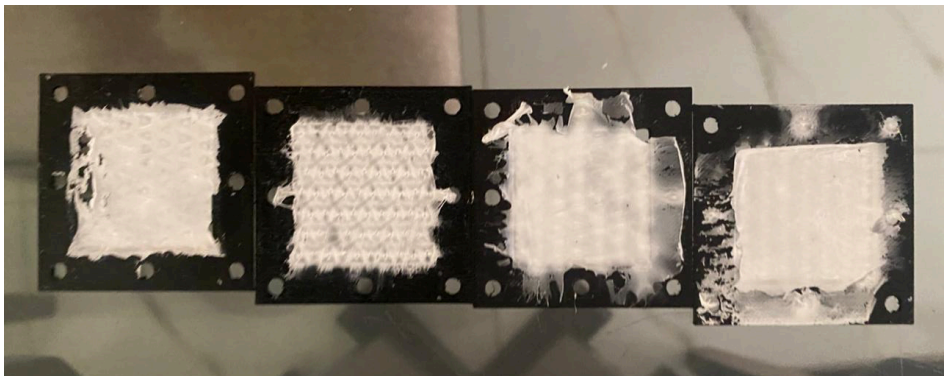


Figure 8. Electrospun matrix on PLA backbone.

The electrospinning solution was also optimised with the ideal polymer wt/v% and ratio of PVA and PEO. The wt/v% was optimised to obtain a solution with an ideal viscosity. The ratio of PVA to PEO was also optimised in order to maintain an ideal solution rheology during electrospinning.

2.3 TiO_2 Concentration Optimization

Rutile TiO_2 particles with a diameter of approximately 20 nm were deposited during the oxide optimization step. The 20 nm diameter was chosen due to its wide-scale availability and economic feasibility. The TiO_2 loading relative to the total amount of dry polymer was varied to 0, 1 and 2 wt%. Each formulation containing TiO_2 was prepared in the same manner before electrospinning to minimise any potential differences from sample preparation. Post-electrospinning, the formed fibres were examined for indicators of fibre quality and the preservation of pore space. The fog collection performance was then determined by testing efficiency in the laboratory fog chamber.

2.4 SiO_2 Concentration Optimization

Amorphous SiO_2, supplied as a 30 wt% colloidal dispersion with a nominal particle diameter of 30 nm, was evaluated as the second inorganic additive at 0, 1 and 3 wt% levels within the combined oxide system. Each formulation was based upon the same polymer composition, solution volume and electrospinning conditions. Since the nanoparticle concentration within the electrospinning solution could impact the resulting nanofibres, the formulations were evaluated both through water collection in the laboratory fog chamber and microscopic examination. The composition carried forward for the final mesh consisted of 1 wt% SiO_2.

2.5 Hydrated Tensile Strength Testing

Tensile testing was performed on the nanofiber matrix in its hydrated state to determine the effect of the colloidal silica on the wet-state mechanical properties of the nanofiber network. Four different types of nanofibers were evaluated:

- PVA/PEO only
- PVA/PEO with 1 wt% TiO_2
- PVA/PEO with 1 wt% TiO_2 and 1 wt% SiO_2
- PVA/PEO with 1 wt% TiO_2 and 3 wt% SiO_2

Specimens of each type were cut to a width of 10 mm and a gauge length of 20 mm. The thickness of each specimen was measured along each specimen, and the average thickness was used to calculate the cross-sectional area of each matrix. The specimens were immersed in deionised water for 30 min at room temperature. After removal, the specimens were gently blotted to remove excess water and subjected to

tensile loading at a constant extension rate of 5 mm min^{-1}. At least five specimens of each type were tested. From the stress-strain curves, the ultimate tensile strength was determined.

2.6 Experimental and Computational Optimisation of Nanofibre Formulations

2.6.1 Combined Nanofiller Concentration Optimization

Following the evaluation of individual components, efforts were made to also evaluate the combination of these two components as the matrix for the nanofibres. Five different compositions of TiO_2 and SiO_2 with a total of between 0 and 3 wt% of oxide loading were prepared:

Formulation	**TiO_2 (wt%)**	**SiO_2 (wt%)**	**Total oxide loading (wt%)**
A	0.0	0.0	0.0
B	1.0	0.0	1.0
C	1.0	1.0	2.0
D	2.0	1.0	3.0

Optimization of the fibre compositions was based on two criteria: the need to provide reproducible collection of water from environments with identical fog exposure, and the need to preserve the nanofibre morphology and ease of electrospinning.

Formulation C (1 wt% TiO_2 + 1 wt% SiO_2; 2 wt% total oxide loading) was therefore selected as the final composition of the EcoFog fibres to be utilised in the characterisation and testing of both the controlled and outdoor fog collection capabilities of the fibres (see Results for supporting data).

2.6.2 Exploratory Gaussian Process Modelling

To investigate whether computational methods could complement conventional experimental optimisation, a preliminary Gaussian Process regression model was developed using 36 fog-chamber measurements. The input variables comprised electrospinning tip-to-collector distance, TiO_2 concentration and SiO_2 concentration, while the response variable was the area-normalised water collection rate.

A Matérn 5/2 covariance kernel was employed to model potentially nonlinear relationships between fabrication conditions and collection performance. Unlike conventional regression methods that produce only point estimates, the Gaussian Process generated both predicted collection rates and associated uncertainty estimates [43,46].

Model performance was assessed using configuration-level cross-validation, with replicate measurements grouped to avoid information leakage between training and evaluation. We additionally explored Expected Improvement as an acquisition function for identifying potentially informative candidate configurations [44,45].

This preliminary computational investigation was intended to evaluate the feasibility of uncertainty-aware experimental optimisation and establish a foundation for future integration of machine learning into EcoFog's fabrication process.

2.7 Glutaraldehyde Crosslinking

The PVA/PEO matrix was stabilized after deposition, using vapour phase crosslinking with glutaraldehyde under a fume hood at all times. Vapour treatment was selected, since aqueous solution treatment can swell uncrosslinked PVA and dissolve the matrix. Thus, vapour crosslinking preserves the

fibre geometry during network formation. 25% (v/v) glutaraldehyde solution acted as the vapour source, and the material was exposed to the vapours in a fume hood for 4 hours. Post-electrospinning, the material was washed with deionised water to remove any residual glutaraldehyde. Each material was then dried for 24 hours at room temperature [42].

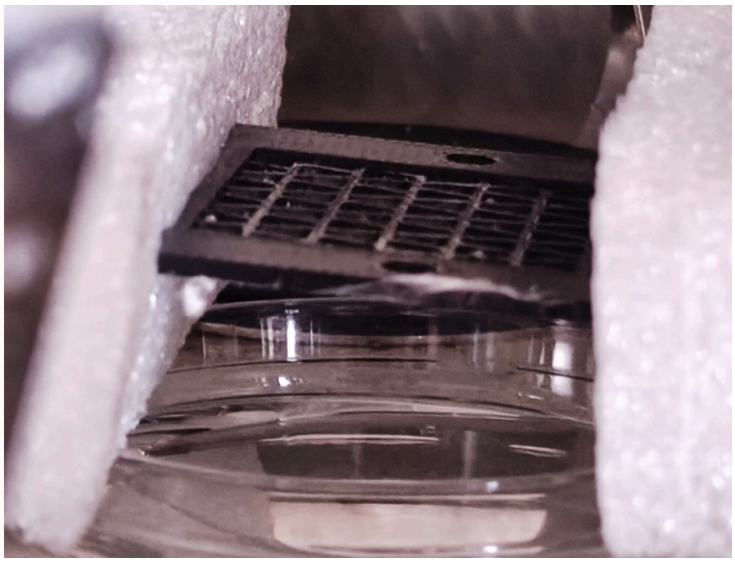


Figure 9. Glutaraldehyde crosslinking of PVA-rich matrix

2.8 Material Characterization

2.8.1 Field-emission Scanning Electron Microscopy (FESEM)

FESEM was conducted to evaluate fibre uniformity, bead formation, and agglomeration of the matrix. FESEM was also conducted for the PLA support to assess surface smoothness and morphology. The images of various loadings of TiO_2 and SiO_2 were observed for agglomerates and fiber uniformity, allowing for the optimization of nanofiller loading.

Binary segmentation of FESEM images was conducted to calculate the two-dimensional void fraction. After processing the images, the surface porosity was calculated using:

$$P_{2D} = \frac{A_{void}}{A_{image}} \times 100\%$$

This quantity was used to characterise the projected void fraction in the analysed micrographs.

2.8.2 Energy Dispersive X-ray Spectroscopy (EDS)

The EDS was used to observe if Ti and Si were still present in the matrix post-crosslinking. Since EDS samples a finite interaction volume and is influenced by surface geometry, the obtained weight percentages were not equated to the bulk formulation of weight percentages.

2.8.3 Fourier-transform Infrared Spectroscopy (FTIR)

FTIR spectra were collected for the matrix before and after crosslinking. The analysis focused on the O-H stretching region and the C-O/C-O-C region. FTIR was conducted to provide further evidence of crosslinking in the matrix through the changes in the functionalised groups before and after crosslinking.

2.8.4 X-ray Diffraction (XRD)

XRD was used to identify the crystalline phase of TiO_2 and confirm the rutile phase of TiO2. The diffraction patterns would also showcase the change in the crystalline structure post crosslinking. The obtained patterns would then be compared with data published in the International Centre for Diffraction Data (ICDD) to identify the phase of TiO_2 and PVA.

2.8.5 Sessile Drop Contact Angle

Sessile-drop contact angles were measured for the PLA backbone, non-crosslinked matrix, and crosslinked matrix. The measurement calculates the static wettability contrast but not droplet mobility itself.

2.9 Fog Water Collection

2.9.1 Fog chamber Design

A closed chamber was designed using a 12-foot by 6-foot by 6-foot rectangular prism. The material used was transparent acrylic sheets and sealed using a synthetic rubber-based adhesive. An ultrasonic humidifier (Kent Dew Humidifier 4L) was used to simulate fog-like conditions inside the chamber. An inlet was carved out in the chamber, and a pipe was attached to allow fog to flow into the system. To simulate airflow, an exhaust fan was used at the end of the chamber.

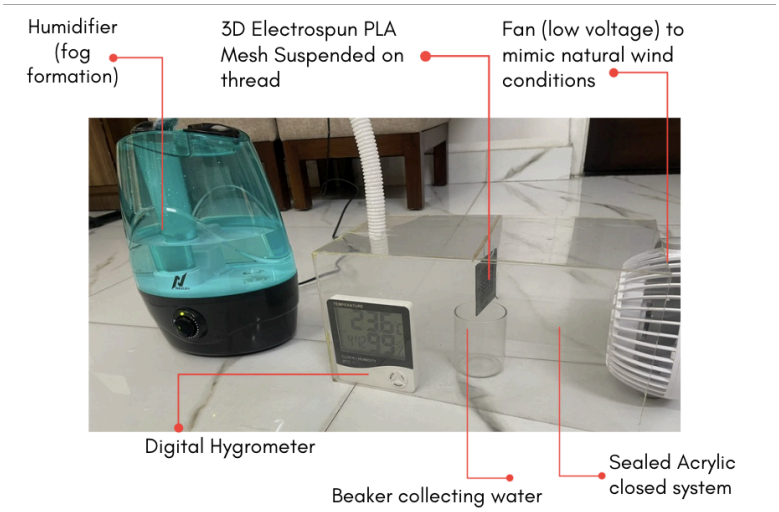


Figure 10. Fog chamber set-up

The specimen position, orientation, exposure time and geometry were kept constant. The chamber conditions were also kept constant with a 3 m s^{-1} airflow, 98-100% relative humidity (RH) and a temperature of 30 ℃.

Collection rate was calculated using:

$$q_A = \frac{m_c}{At}$$

Where m_c is the mass of water collected, A is the projected area of the specimen, and t is the exposure time.

2.9.2 Outdoor Testing

S1, S2, and S3 were mounted simultaneously over five consecutive days, near Mount Abu, Rajasthan, India. All three materials experienced the same fog intensity, wind speed, and temperature (10—20℃), allowing environmental variability to be treated as a matched factor rather than uncontrolled.

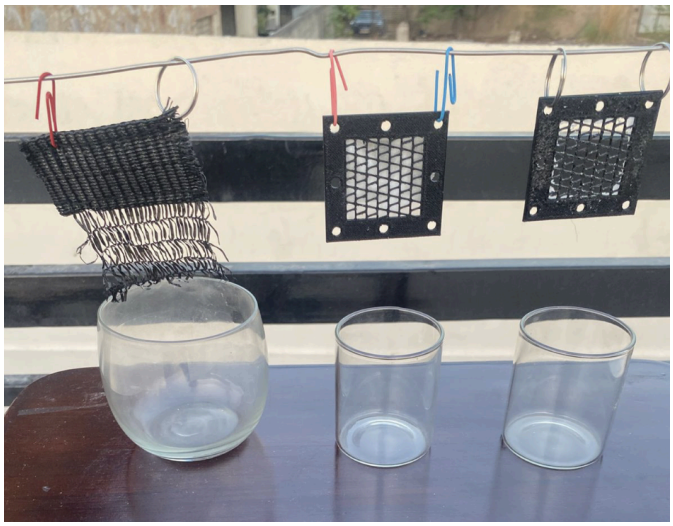


Figure 11. Outdoor testing set-up for S1, S2, and S3

The collection rate was calculated in the same manner. Retained water by the material was estimated using:

$$R = m_{wet} - m_{dry}$$

Where m_{wet} is the mass of the material after interception with fog, and m_{dry} is the mass of the material before exposure to fog or moisture.

3. Results

3.1 Material Characterization

3.1.1 Field Emission Scanning Electron Microscopy (FESEM)

The PLA backbone showcases a homogeneous structure with smooth surfaces. The FESEM proves structural robustness but poor wetting capability, thereby fulfilling the function of the PLA as a support.

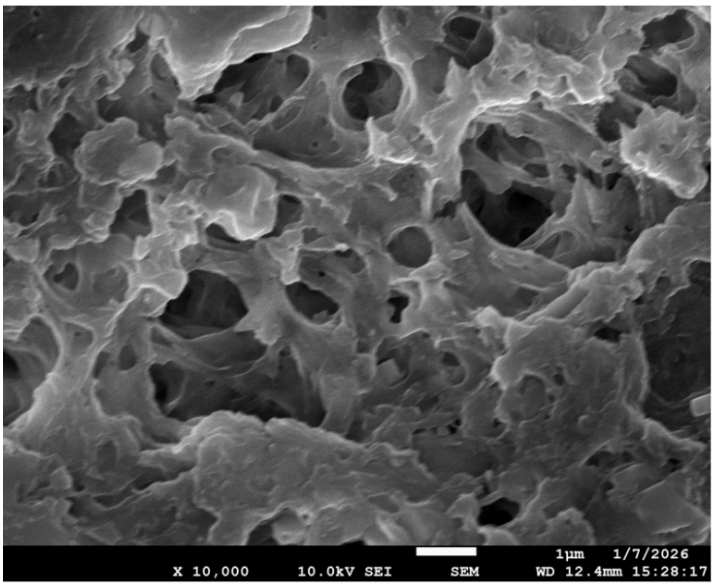


Figure 12. SEM of PLA backbone

The FESEM of the crosslinked matrix showcases a uniform nanofiber diameter which is bead-free. Moreover, very few agglomerates are present compared to the higher-loading formulations. Image segmentation yielded a local two-dimensional porosity of 48.60 $\pm$ 3.89% across four analysed fields. The

reported 95% confidence interval is 42.40-54.80%, and the coefficient of variation is 8.01%. The relatively small coefficient of variation showcases uniform deposition during electrospinning.

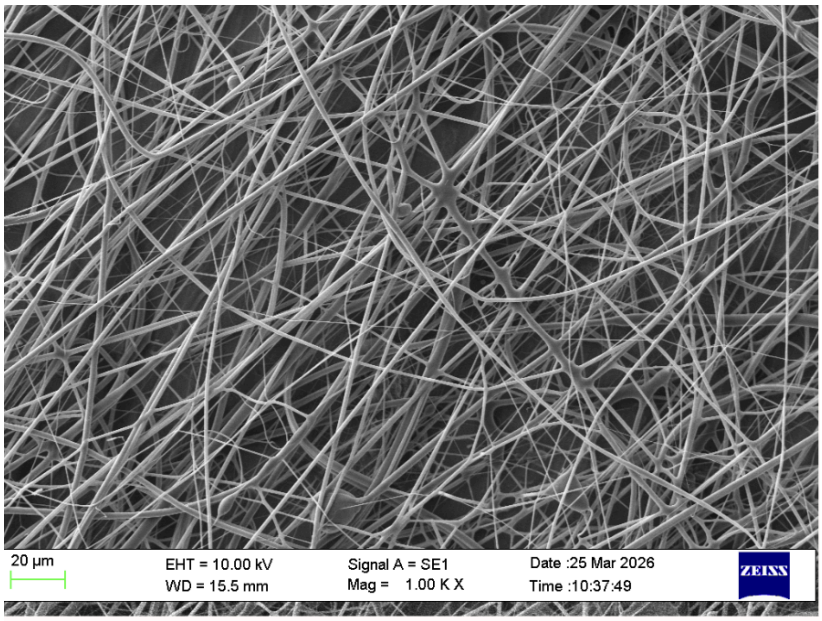


Figure 13. SEM of Optimised matrix (1 wt% TiO_2 and 1 wt% SiO_2)

The rejected formulation (2 wt% TiO_2 and 1 wt% SiO_2) showcased larger non-uniformity and dense regions. Therefore, the FESEM provided morphological evidence for why additional oxide would hinder uniformity of the matrix. Nanoparticle agglomerates alter the conductivity and rheology of the electrospinning solution, affecting the electrospinning jet and Taylor cone formation. The dense regions also hinder adequate pore size and surface area.

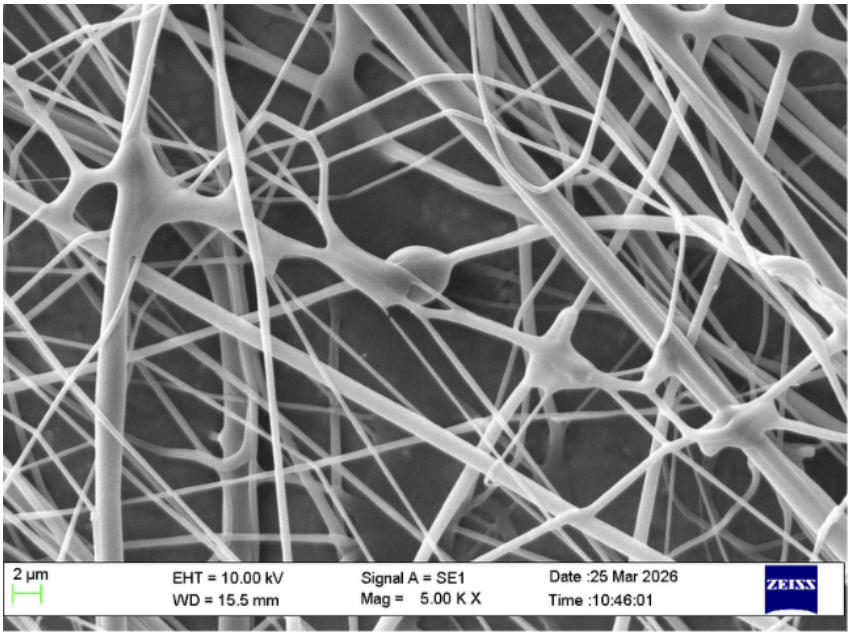


Figure 14. SEM of matrix with (2 wt% TiO_2 and 1 wt% SiO_2) with clear bead formation

3.1.2 Contact Angle

Static contact angle measurements showed angles of approximately 100.0° for the PLA backbone, 29.3° for the non-crosslinked electrospun matrix and 22.4° for the crosslinked matrix. The significant difference between the PLA backbone and the crosslinked matrix establishes the intended dual wettability architecture.

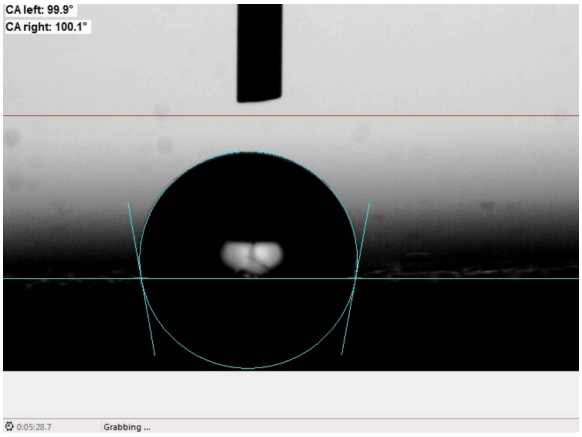


Figure 15. Contact angle measurement for PLA backbone

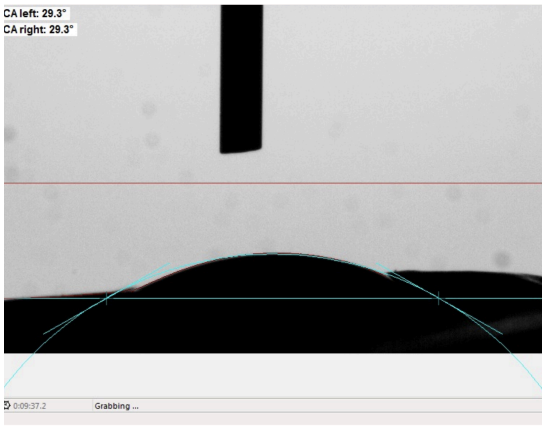


Figure 16. Contact angle measurement for non-crosslinked electrospun surface

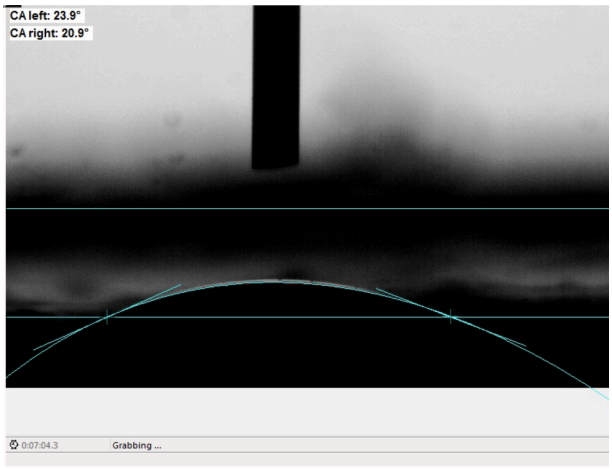


Figure 16. Contact angle measurement for crosslinked electrospun surface

The matrix remained strongly wettable post-crosslinking. This was not guaranteed because crosslinking consumes some of the PVA hydroxyl groups. The methodology therefore achieved resistance to dissolution in aqueous conditions without sacrificing wettability.

The contrasting wettability range between the nanofiber matrix and the PLA backbone supports the intended role of each component. The nanofiber matrix favours droplet pinning after impact, which helps collect large amounts of water. Once the droplet interacts with the hydrophobic PLA backbone, it is transported into a collection gutter.

3.1.3 FTIR Analysis of Polymer Network

The non-crosslinked and crosslinked matrix spectra displayed several differences. The broad O-H stretching envelope shifted from approximately 3350 to 3290 cm^{-1}. The aliphatic C-H region remained near 2929 cm^{-1}. The carbonyl peak near 1755 cm^{-1} before crosslinking was reduced afterwards; the C-O/ C-O-C region (1000-1150 cm^{-1}) also became sharper post-electrospinning.

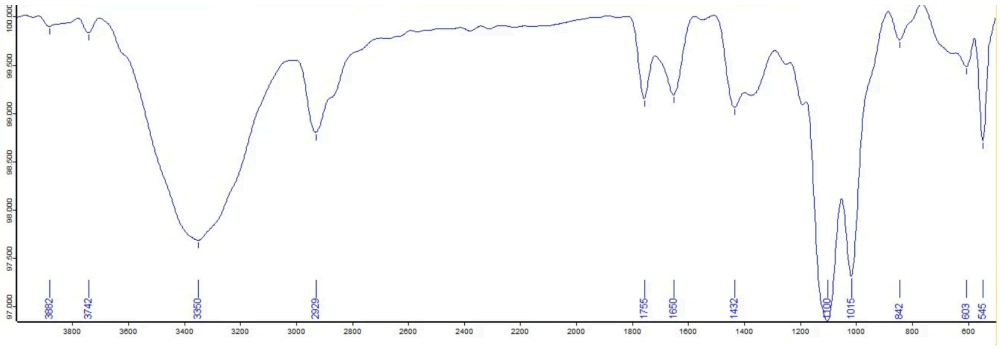


Figure 17. FTIR Spectra for non -crosslinked electrospun matrix

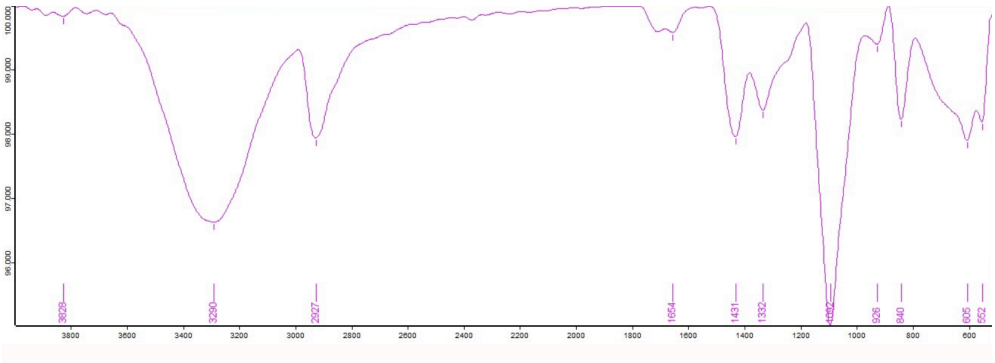


Figure 18. FTIR Spectra for crosslinked electrospun matrix

These observations are consistent with the treatment of the PVA network post-electrospinning. The reduced carbonyl feature showcases a reduction in the residual acetate species. Acid-catalysed hydrolysis of residual poly(vinyl acetate) could serve as an explanation for the decrease in the contact angle post-crosslinking, since the acetate carbonyl disappears while the O-H region changes concurrently. The

spectral changes support the chemical stabilisation of the matrix, while the contact angle measurement reveals the retained hydrophilicity after crosslinking.

3.1.4 Energy-dispersive X-ray Spectroscopy (EDS)

EDS of the crosslinked matrix gave the following composition:

Element	Weight %	Atomic %	Error %
C K	82.5	87.5	7.3
O K	14.4	11.4	10.6
Na K	0.5	0.3	9.8
Si K	0.7	0.3	6.7
Ti K	1.8	0.5	10.5

Figure 19. EDS composition table for crosslinked matrix

The Ti and Si signals show that both TiO_2 and SiO_2 remain detectable after crosslinking and washing. The EDS weight fractions are not directly comparable with the bulk formulation wt% since the interaction volume is local and oxide additives may be enriched near the surface of the fibre. The low Na signal is important, as the backbone was pretreated with an NaCl bath for improved conductivity.

3.1.5 X-ray Diffraction (XRD)

The non-crosslinked matrix exhibited a broad semi-crystalline feature in the PVA -associated reflection (2θ: 19-20°). The TiO_2 reflections at 27.4°, representing plane (110), showcased by the third peak in the non-crosslinked diffractogram, are consistent with the rutile phase according to ICDD Card No. 21-1276 and in agreement with the material specification used in the study.

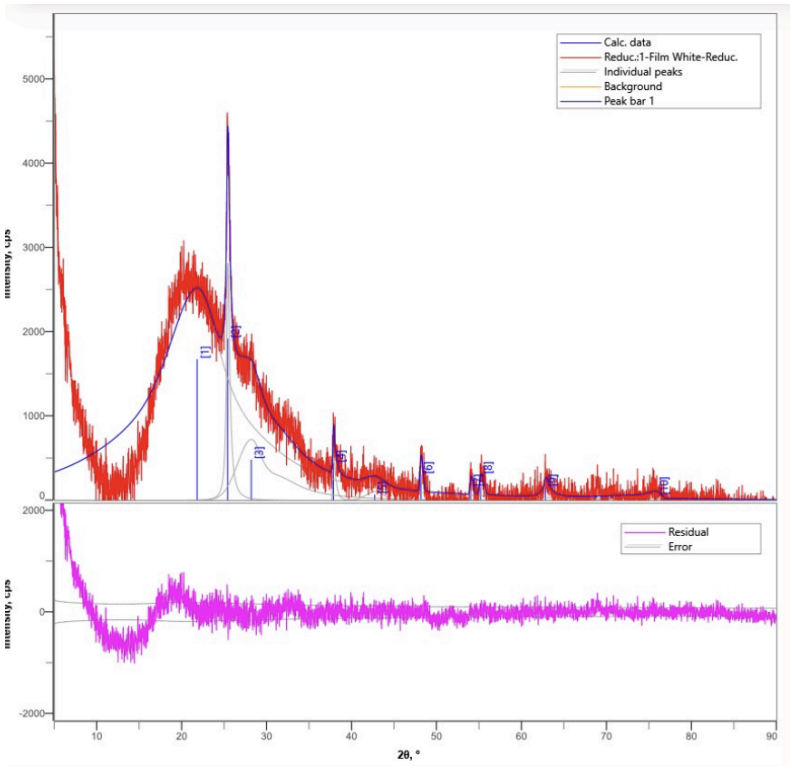


Figure 20. Diffraction pattern for non-crosslinked electrospun surface

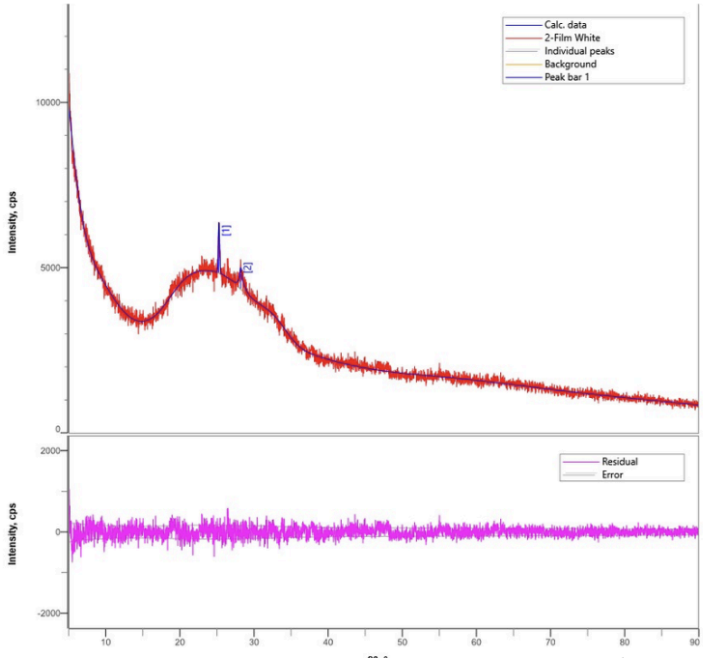


Figure 21. Diffraction pattern for crosslinked electrospun surface

Post- crosslinking, the pattern becomes broader and more curved, forming a broad amorphous halo. This is consistent with a reduction in polymer packing due to a newly formed covalently constrained network.

3.2 Multiscale Optimisation of Collector Architecture

3.2.1 Support Geometry and Shade Coefficient

The geometry of the support structure influenced the projected performance of the collectors. The trapezoidal lattice architecture exhibited the highest predicted collection rate of 15.6 ± 0.8 mg cm^{-2} h^{-1}. The downward-converging cells of the trapezoidal lattice allowed for air passage while permitting drainage of the collected water.

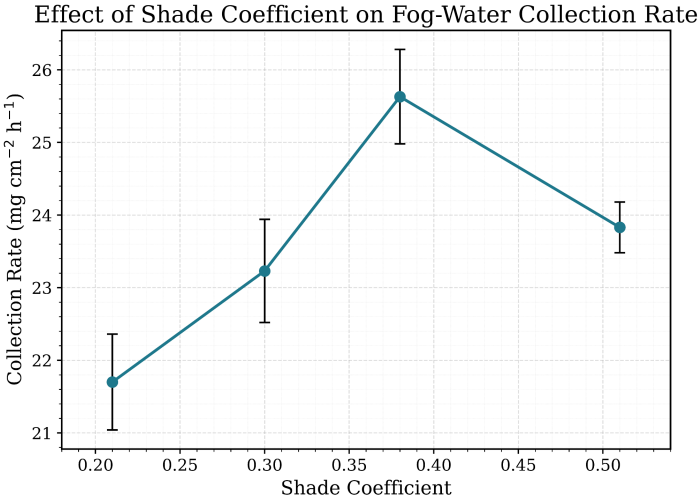


Figure 22. Effect of support shade coefficient on fog-water collection rate.

The dry shade coefficient of this support structure was calculated as: $SC = 0.388$.

3.2.2 Polymer Concentration and PVA/PEO Ratio

The concentration of the polymer had a considerable effect upon fibre formation; too low a concentration results in insufficient entanglement of the polymer chains to form fibres, but an excessively high concentration results in the formation of thick fibres due to excessive viscosity. The optimum for PEO

was similar. The ratio of 90:10 PVA: PEO resulted in the most uniform network and highest collection rates within the tested ratios. A 90:10 ratio of PVA: PEO proved to be the best balance between the required jet stability and fibre integrity. Although PEO can increase the entanglement of PVA chains, excessive amounts can lead to softer deposits. Thus, this ratio was selected for use in the following experiments.

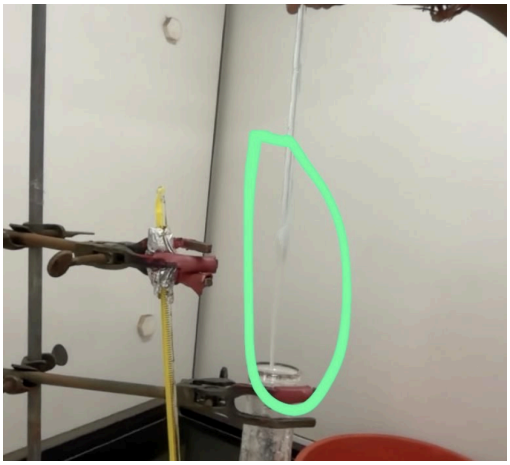


Figure 23. Failed concentration of 22 wt%, which led to excessive viscosity

3.2.3 Spinning Distance

The tip-to-collector distance had an impact on both jet stretching and solvent evaporation. Spinning at a distance of 5.0 cm resulted in insufficient drying of the spun fibres, which led to fused and beaded fibres. At 10.0 cm spinning distances, the collected fibres became more diffuse across the substrate. Thus, 7.5 cm spinning distances resulted in the most homogeneous networks and the highest collection rate of 51.2 ± 2.1 mg cm^{-2} h^{-1}.

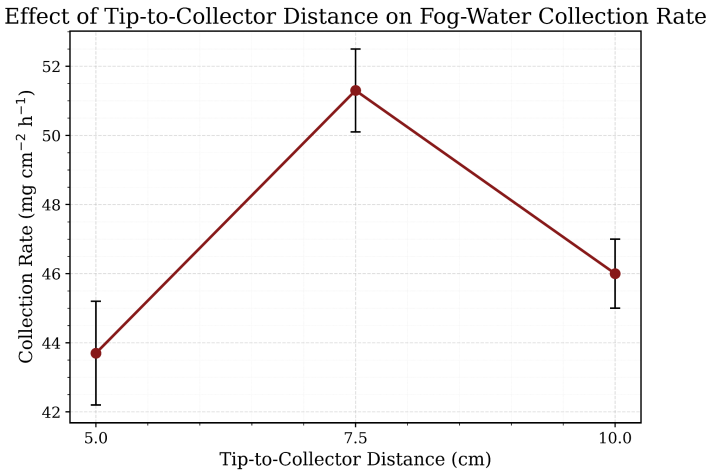


Figure 24. Effect of electrospinning tip-to-collector distance on fog-water collection rate.

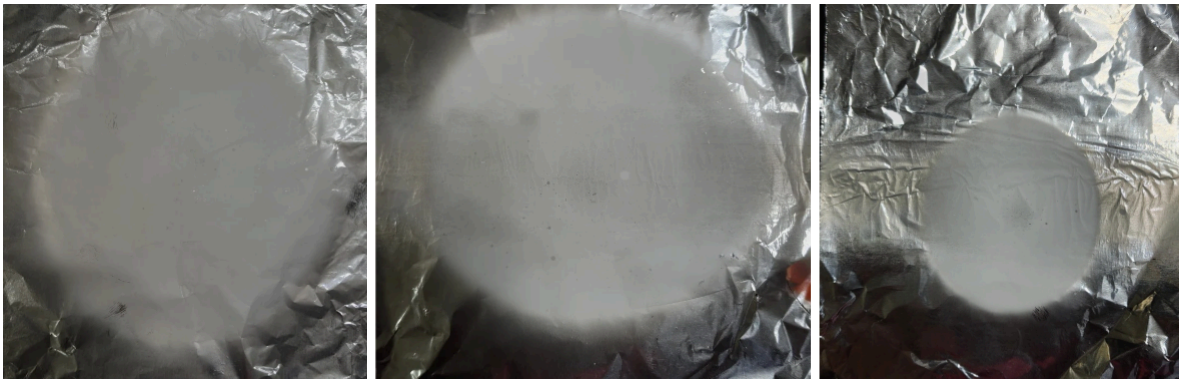


Figure 25. Effect of increasing tip-to-collector distance on deposition (qualitative) (5.0 cm, 7.5 cm, and 10.5 cm, respectively)

The 7.5 cm distance was therefore fixed before oxide incorporation.

3.2.4 TiO_2 Loading

The inclusion of oxides led to a non-monotonic response in collection rates. The addition of 1 wt% TiO_2 led to the largest increase in the collection rate. This was due to the formation of polar Ti-OH groups that could interact with both water and PVA nanofibres containing hydroxyl groups, enhancing the impaction

and retention of micron-scale water droplets. At higher concentrations, the open fibre network was disrupted, resulting in lower retention of water droplets.

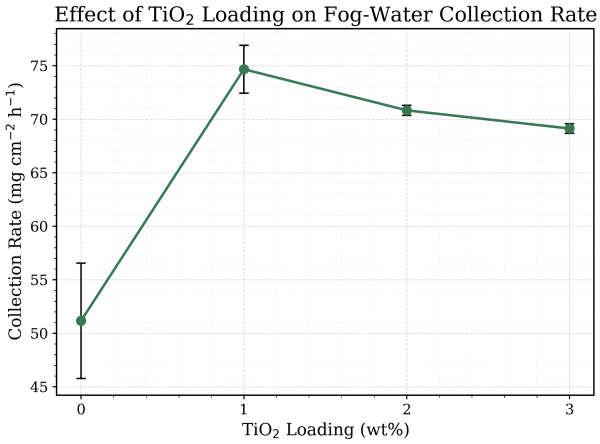


Figure 26. Effect of TiO_2 loading on fog-water collection rate of the PVA/PEO nanofibre network.

3.2.5 SiO_2 Loading

The loading of SiO_2 gave rise to a non-monotonic response in the mechanical properties of the network. The force that could be sustained by the oxide-free PVA/PEO network was as high as 22.18 ± 1.26 N; at 1 wt% SiO_2 loading, this increased to 29.80 ± 2.07 N, a 34.3% increase in the strength of the network. Beyond 1 wt%, however, the strength of the network decreased. Thus, it appears that while 1 wt% SiO_2 acted to improve the load transfer across the network, at higher loadings the increased defects and particle association counteracted its benefits.

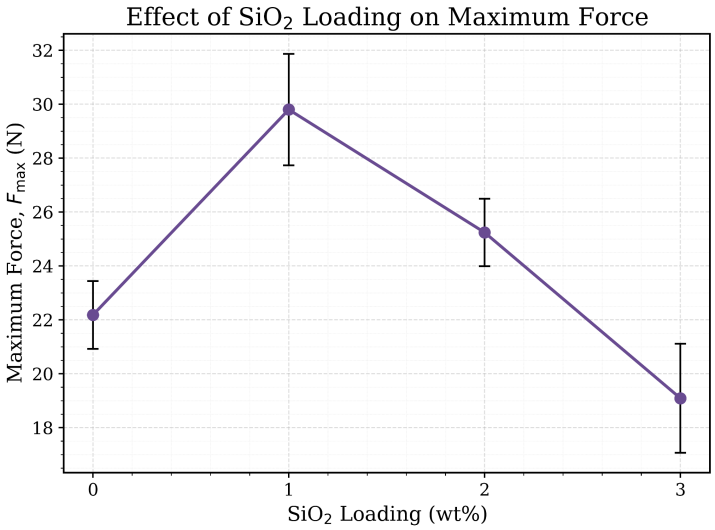


Figure 27. Effect of SiO_2 loading on the maximum force sustained by the PVA/PEO nanofibre network.

3.2.6 Total Oxide Loading and Final Formulation

The addition of TiO_2 produced a non-monotonic response (Table 6). 1 wt% TiO_2 produced the optimum collection performance. At these low loadings, the hydroxylated Ti-OH groups can interact with the water and the hydroxyl groups of the PVA. Further additions of TiO_2 and SiO_2 led to decreased collection efficiency, likely due to the increased agglomeration and reduced fibre uniformity.

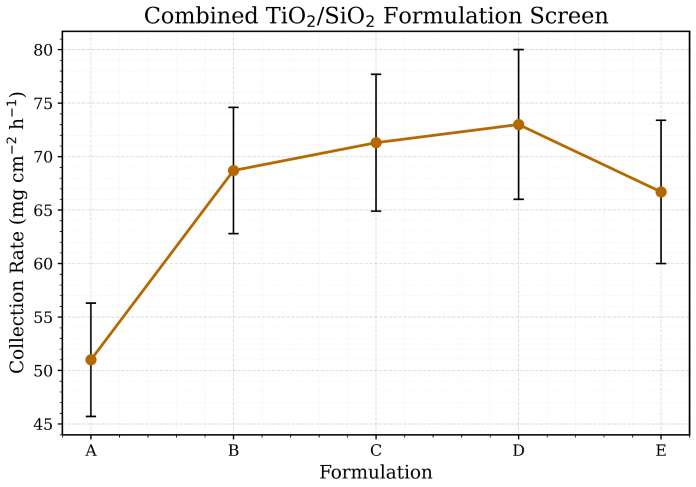


Figure 28. Fog-water collection rate across the combined TiO_2/SiO_2 formulation series.

For these reasons, formulation C containing 1 wt% TiO_2 and 1 wt% SiO_2 was chosen as the final composition for EcoFog.

3.2.7 Preliminary Computational Optimisation

The exploratory Gaussian Process analysis demonstrated both the potential and limitations of modelling fog-water collection from a relatively small experimental dataset. Configuration-level cross-validation yielded a mean absolute error of 9.86 mg cm^{-2} h^{-1} and a root mean squared error of 11.99 mg cm^{-2} h^{-1}. Nominal 95% predictive intervals contained 69.4% of the held-out observations, indicating that the model underestimated uncertainty in parts of the experimental design space.

Despite these limitations, the analysis provided an initial framework for examining fabrication parameters through probabilistic predictions rather than relying exclusively on sequential experimental comparisons. The exploration of Expected Improvement further illustrated how model uncertainty could be incorporated into the prioritisation of subsequent experiments.

These findings suggest that Gaussian Process-based Bayesian optimisation may become useful for identifying promising material configurations, particularly as additional experimental observations improve model accuracy and uncertainty calibration.

3.3 Controlled Fog-chamber Performance

The PVA/PEO network significantly contributed to the increase in fog collection rates before the introduction of either nanofiller (Table 9).

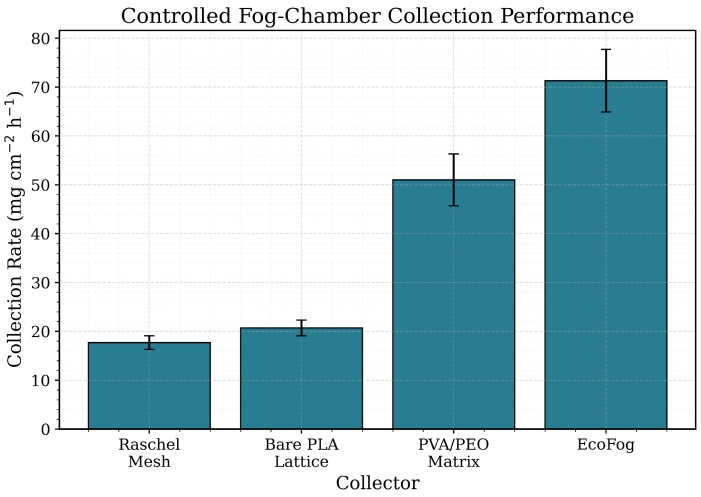


Figure 29. Controlled fog-chamber collection performance of Raschel mesh, bare PLA lattice, oxide-free PVA/PEO nanofibre matrix and EcoFog.

The similar rates of fog collection between the Raschel mesh and the bare PLA lattice suggest that the geometric optimization, though advantageous, is not the main contributing factor to the increase in collection rates. Instead, the increase in collection rates with the incorporation of the nanofibre network is likely due to the increase in the fine fibres that can intercept the fog droplets.

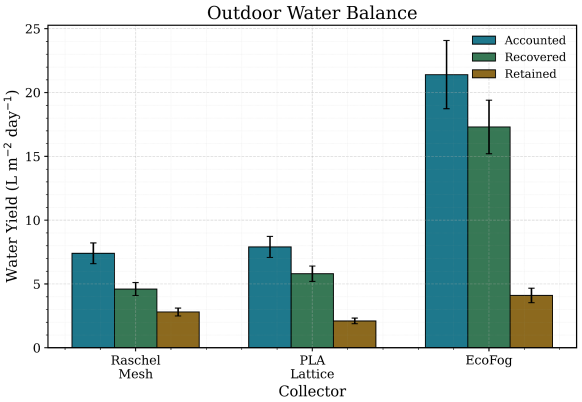
Since fog entered the chamber in droplet form, the results support the role of interception and impaction in the collection of fog. The remaining processes, namely adhesion, spreading, coalescence, and drainage, apply to the droplets that land on the surface.

3.4 Outdoor Validation and Water Recovery

3.4.1 Field Performance

Open-field testing was conducted at Mount Abu, Rajasthan, India (approx. 1,722 m a.s.l.). During fog-exposure periods, the environmental conditions included temperatures of approximately 22-25 °C, relative humidity of 90-100%, and wind speeds of 2.2-4.8 m s^{-1}. EcoFog, compared alongside PLA lattice structure and commercially used Raschel mesh samples, were mounted onto the same frames such that all

samples had similar projected areas and orientations. Each sample was exposed to the outdoors over a five-day period. Water collected from each sample was measured at 12-hour intervals. The daily water yield was calculated by dividing the collected water by the projected area and exposure time.



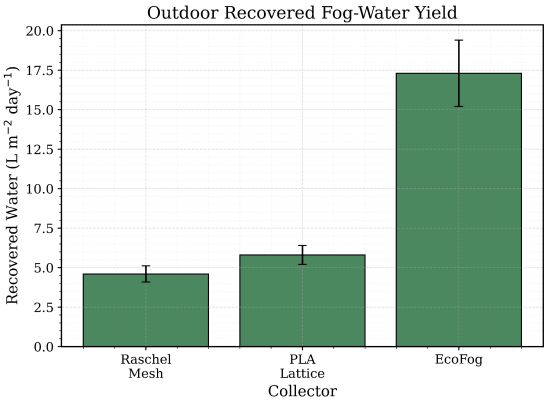


Figure 30. Five-day outdoor water balance Raschel mesh, PLA lattice, and EcoFog

Figure 31. Recovered fog-water yield during simultaneous outdoor testing

Outdoor validation of the water recovery capabilities of EcoFog is consistent with results obtained indoors. While modification of the support geometry alone provides some improvement in water recovery, the addition of the nanofibres and nanoparticles contributes the majority of the improvement in recovery rate.

The greater total water collection of EcoFog did not, however, yield a lower recovery fraction, showcasing that coplanar heterowetting architecture performs as intended: first droplets are attracted through the hydrophilic matrix, aided by TiO_2 nanofillers; then, coalescence into larger droplets takes place across the fibre network; finally, as droplets move to the hydrophobic PLA backbone, they are effectively shed off without bridging the pores.

3.5 Limitations

The conclusions are bounded by the duration and environmental factors of the experiments. Outdoor testing was conducted at one site. The fog simulation chamber had limited airflow range, temperature, and relative humidity fluctuations. The measured 3.76-fold ratio therefore establishes the comparative performance under the specified conditions, though the outdoor testing presented fluctuations. Long-duration exposure is also needed to quantify the UV degradation, fouling, wet-dry cycling and changes in the crosslinked nanofiber matrix.

Certain quantities are yet to be measured: advancing and receding contact angles to quantify the contact angle hysteresis and droplet depinning; pressure drop measurements to resolve the aerodynamic cost of the matrix. Moreover, a complete leaching analysis of collected water is required before any claims can be made about the potability of water without downstream treatment.

A further direction for EcoFog is the development of a fully integrated machine-learning-guided optimisation framework. While the preliminary Gaussian Process analysis demonstrated the feasibility of modelling fabrication parameters and their uncertainties, its predictive reliability was constrained by the limited number of distinct experimental configurations.

Future work will expand the dataset through systematically selected combinations of spinning distance, nanoparticle concentrations and additional fabrication parameters. Bayesian acquisition functions could then be incorporated into a closed-loop experimental framework, in which each new physical measurement updates the predictive model and informs the selection of subsequent experiments. Such an approach could reduce the number of laboratory trials required to identify high-performing formulations.

With additional measurements of aerodynamic pressure drop, mechanical durability and drainage efficiency, the framework could also be extended towards multi-objective Bayesian optimisation,

allowing material configurations to be evaluated against several competing performance criteria simultaneously.

4. Conclusion

Ecofog was created to address the current limitations in passive fog harvesting, which include field-deployable meshes that are scalable. Many lab-based solutions often involve fabrication complexity, niche materials, or complex processes. Ecofog demonstrates that these constraints can be overcome with a coplanar architecture that allows for simultaneous capture and drainage.

The final collector combines a PLA backbone with a crosslinked electrospun PVA/PEO matrix containing low loadings of TiO_2 and SiO_2. The geometry of the PLA surface was optimised to a calculated shade coefficient of approximately 0.388. The electrospinning dope was optimised to an 18 wt% solution and a 90:10 PVA: PEO mass ratio. The nanofiller loading was then screened afterwards, and a 1 wt% TiO_2 and 1 wt% SiO_2 formulation created an open and uniform morphology. Higher loadings produced increasingly non-uniform fibres with larger agglomerated regions.

Characterisation confirmed the intended wettability contrast and morphology. The optimised matrix retained around a 48.6 ± 3.9% two-dimensional porosity across the analysed FESEM fields. EDS detected both Ti and Si after crosslinking and washing. FTIR showcased consistent formation of stabilised acetal-containing PVA networks. XRD confirmed the rutile phase of TiO_2, the semi-crystalline structure of the non-crosslinked matrix and the amorphous pattern of the crosslinked matrix. The static contact angle testing showcased the strong contrast between the nanofiber matrix and PLA support.

Performance data highlighted that the nanofiber matrix is the dominant source of fog water collection, with oxide functionalisation. Under simultaneous five-day outdoor testing, EcoFog collected 17.3 ± 2.10 L m^{-2} day^{-1}, compared with 4.6 ± 0.51 L m^{-2} day^{-1} for the conventional Raschel mesh. Thus, EcoFog collected 3.76 times more than the Raschel mesh. The gain cannot be attributed to the PLA support since

there was no significant improvement, showing that the improvements were not a result of shape change. The field testing further provides evidence of greater capture, while showcasing improved drainage.

Our preliminary exploration of Gaussian Process regression also revealed opportunities for integrating machine learning into the experimental design process. Future development of a closed-loop Bayesian optimisation framework could enable more efficient exploration of interacting fabrication parameters, extending EcoFog's optimisation beyond conventional sequential experimentation.

The optimisation establishes the conditions under which a scalable, passive, fluorine-free system achieves high water collection, whilst retaining the properties that make it effective during field deployment. Fuse deposition modelling and single-jet electrospinning were used as part of a lab methodology, but can be scaled with moulded thermoplastic panels and industrial needleless electrospinning. PLA is also treated as an industrially compostable polymer, making the system compostable after its lifetime use.

5. References

[1] World Health Organisation; UNICEF. Progress on Household Drinking Water and Sanitation 2000-2024: Special Focus on Inequalities; WHO/UNICEF Joint Monitoring Programme, 2025.

[2] Klemm, O.; Schemenauer, R. S.; Lummerich, A.; et al. Fog as a fresh-water resource: overview and perspectives. Ambio 2012, 41, 221-234. doi:10.1007/s13280-012-0247-8

[3] Schemenauer, R. S.; Cereceda, P. A proposed standard fog collector for use in high-elevation regions. J. Appl. Meteorol. 1994, 33(11), 1313-1322. doi:10.1175/1520-0450(1994)033<1313:APSFCF>2.0.CO;2

[4] Kennedy, B. S.; Boreyko, J. B. Bio-inspired fog harvesting meshes: a review. Adv. Funct. Mater. 2024, 2306162. doi:10.1002/adfm.202306162

[5] Verbrugghe, N. Water harvesting through fog collectors. Int. J. Low-Carbon Technol. 2023, 18, 356-373. doi:10.1093/ijlct/ctac129

[6] Rivera, J. de D. Aerodynamic collection efficiency of fog water collectors. Atmos. Res. 2011, 102(3), 335-342. doi:10.1016/j.atmosres.2011.08.005

[7] Fernandez, D. M.; et al. Fog water collection effectiveness: mesh intercomparisons. Aerosol Air Qual. Res. 2018, 18(1), 270-283. doi:10.4209/aaqr.2017.01.0040

[8] Azeem, M.; et al. Optimal design of multilayer fog collectors. ACS Appl. Mater. Interfaces 2020, 12(6), 7736-7743. doi:10.1021/acsami.9b19727

[9] Knapczyk-Korczak, J.; Szewczyk, P. K.; Ura, D. P.; Berent, K.; Stachewicz, U. Hydrophilic nanofibers in fog collectors for increased water harvesting efficiency. RSC Adv. 2020, 10, 22335-22342. doi:10.1039/D0RA03939J

[10] Lee, J. H.; et al. Unclogged Janus mesh for fog harvesting. ACS Appl. Mater. Interfaces 2022, 14(18), 21713-21726. doi:10.1021/acsami.2c03419

[11] Li, J.; et al. Aerodynamics-assisted, efficient and scalable kirigami fog collectors. Nat. Commun. 2021, 12, 5484. doi:10.1038/s41467-021-25764-4

[12] Damak, M.; Varanasi, K. K. Electrostatically driven fog collection using space charge injection. Sci. Adv. 2018, 4(6), eaao5323. doi:10.1126/sciadv.aao5323

[13] Quéré, D. Wetting and roughness. Annu. Rev. Mater. Res. 2008, 38, 71-99. doi:10.1146/annurev.matsci.38.060407.132434

[14] Seo, D.; Lee, J.; Lee, C.; Nam, Y. The effects of surface wettability on fog and dew moisture harvesting on tubular surfaces. Sci. Rep. 2016, 6, 24276. doi:10.1038/srep24276

[15] Xue, J.; Wu, T.; Dai, Y.; Xia, Y. Electrospinning and electrospun nanofibers: methods, materials, and applications. Chem. Rev. 2019, 119(8), 5298-5415. doi:10.1021/acs.chemrev.8b00593

[16] Mansur, H. S.; et al. FTIR characterization of PVA hydrogel chemically crosslinked with glutaraldehyde. Mater. Sci. Eng. C 2008, 28(4), 539-548. doi:10.1016/j.msec.2007.10.088

[17] Shaikh, R. P.; et al. Crosslinked electrospun polyvinyl alcohol nanofibrous membranes. Biofabrication 2012, 4(2), 025002. doi:10.1088/1758-5082/4/2/025002

[18] Bolto, B.; Tran, T.; Hoang, M.; Xie, Z. Crosslinked poly(vinyl alcohol) membranes. Prog. Polym. Sci. 2009, 34(9), 969-981. doi:10.1016/j.progpolymsci.2009.05.003

[19] Zhuravlev, L. T. The surface chemistry of amorphous silica: the Zhuravlev model. Colloids Surf. A 2000, 173(1-3), 1-38. doi:10.1016/S0927-7757(00)00556-2

[20] Lettieri, S.; et al. Charge carrier processes and optical properties in TiO_2 and TiO_2-based heterojunction photocatalysts. Materials 2021, 14(7), 1645. doi:10.3390/ma14071645

[21] Avramescu, M.-L.; Rasmussen, P. E.; Chénier, M.; Gardner, H. D. Influence of pH, particle size and crystal form on dissolution behaviour of engineered nanomaterials. Environ. Sci. Pollut. Res. 2017, 24, 1553-1564. doi:10.1007/s11356-016-7932-2

[22] OECD. Reconciling Terminology of the Universe of Per- and Polyfluoroalkyl Substances; OECD Series on Risk Management No. 61; Paris, 2021.

[23] Persano, L.; Camposeo, A.; Tekmen, C.; Pisignano, D. Industrial upscaling of electrospinning and applications of polymer nanofibers. Macromol. Mater. Eng. 2013, 298(5), 504-520. doi:10.1002/mame.201200290

[24] Chamas, A.; et al. Degradation rates of plastics in the environment. ACS Sustainable Chem. Eng. 2020, 8(9), 3494-3511. doi:10.1021/acssuschemeng.9b06635

[25] Yu, Z.; et al. Desert beetle-inspired superwettable patterned surfaces for water harvesting. Small 2017, 13(36), 1701403. doi:10.1002/smll.201701403

[26] Yang, X.; et al. A twice electrochemical-etching method to fabricate superhydrophobic-superhydrophilic patterns for biomimetic fog harvest. Sci. Rep. 2017, 7, 8816. doi:10.1038/s41598-017-09108-1

[27] Fahmi, A.; Minot, C. A theoretical investigation of water adsorption on titanium dioxide surfaces. Surf. Sci. 1994, 304(3), 343-359. doi:10.1016/0039-6028(94)91345-5

[28] Farah, S.; Anderson, D. G.; Langer, R. Physical and mechanical properties of PLA, and their functions in widespread applications — a comprehensive review. Adv. Drug Deliv. Rev. 2016, 107, 367-392. doi:10.1016/j.addr.2016.06.012

[29] Castro-Aguirre, E.; Iñiguez-Franco, F.; Samsudin, H.; Fang, X.; Auras, R. Poly(lactic acid) — mass production, processing, industrial applications, and end of life. Adv. Drug Deliv. Rev. 2016, 107, 333-366. doi:10.1016/j.addr.2016.03.010

[30] Woodruff, M. A.; Hutmacher, D. W. The return of a forgotten polymer — polycaprolactone in the 21st century. Prog. Polym. Sci. 2010, 35(10), 1217-1256. doi:10.1016/j.progpolymsci.2010.04.002

[31] Carneiro, O. S.; Silva, A. F.; Gomes, R. Fused deposition modeling with polypropylene. Mater. Des. 2015, 83, 768-776. doi:10.1016/j.matdes.2015.06.053

[32] Aslam, M.; Kalyar, M. A.; Raza, Z. A. Polyvinyl alcohol: a review of research status and use of polyvinyl alcohol based nanocomposites. Polym. Eng. Sci. 2018, 58(12), 2119-2132. doi:10.1002/pen.24855

[33] Sawatari, C.; Kondo, T. Interchain hydrogen bonds in blend films of poly(vinyl alcohol) and its derivatives with poly(ethylene oxide). Macromolecules 1999, 32(6), 1949-1955. doi:10.1021/ma980900o

[34] Shenoy, S. L.; Bates, W. D.; Frisch, H. L.; Wnek, G. E. Role of chain entanglements on fiber formation during electrospinning of polymer solutions. Polymer 2005, 46(10), 3372-3384. doi:10.1016/j.polymer.2005.03.011

[35] Fong, H.; Chun, I.; Reneker, D. H. Beaded nanofibers formed during electrospinning. Polymer 1999, 40(16), 4585-4592. doi:10.1016/S0032-3861(99)00068-3

[36] Diebold, U. The surface science of titanium dioxide. Surf. Sci. Rep. 2003, 48(5-8), 53-229. doi:10.1016/S0167-5729(02)00100-0

[37] Mui, J.; Ngo, J.; Kim, B. Aggregation and colloidal stability of commercially available Al_2O_3 nanoparticles in aqueous environments. Nanomaterials 2016, 6(5), 90. doi:10.3390/nano6050090

[38] Pirzada, T.; Arvidson, S. A.; Saquing, C. D.; Shah, S. S.; Khan, S. A. Hybrid silica-PVA nanofibers via sol-gel electrospinning. Langmuir 2012, 28(13), 5834-5844. doi:10.1021/la300049j

[39] Park, K.-C.; Chhatre, S. S.; Srinivasan, S.; Cohen, R. E.; McKinley, G. H. Optimal design of permeable fibre network structures for fog harvesting. Langmuir 2013, 29(43), 13269-13277. doi:10.1021/la402409f

[40] J. Long, A. V. Nand, C. Bunt, and A. Seyfoddin, “Controlled release of dexamethasone from poly(vinyl alcohol) hydrogel,” *Pharmaceutical Development and Technology*, vol. 24, no. 7, pp. 839–848, Apr. 2019, doi: 10.1080/10837450.2019.1602632.

[41] A. A. Elshennawy, M. Y. Abdelaal, A. M. Hamed, and M. M. Awad, “Evaluating mesh geometry and shade coefficient for fog harvesting collectors,” *Water Resources Management*, vol. 37, no. 15, pp. 6107–6126, Nov. 2023, doi: 10.1007/s11269-023-03644-4.

[42] A. G. Destaye, C.-K. Lin, and C.-K. Lee, “Glutaraldehyde Vapor Cross-linked Nanofibrous PVA Mat with in Situ Formed Silver Nanoparticles,” *ACS Applied Materials & Interfaces*, vol. 5, no. 11, pp. 4745–4752, May 2013, doi: 10.1021/am401730x.

[43] Rasmussen, C. E.; Williams, C. K. I. Gaussian Processes for Machine Learning. MIT Press, Cambridge, MA, 2006. https://gaussianprocess.org/gpml/

[44] Jones, D. R.; Schonlau, M.; Welch, W. J. Efficient Global Optimization of Expensive Black-Box Functions. Journal of Global Optimization 1998, 13, 455–492. https://doi.org/10.1023/A:1008306431147

[45] Frazier, P. I. A Tutorial on Bayesian Optimization. arXiv 2018, arXiv:1807.02811. https://doi.org/10.48550/arXiv.1807.02811

[46] Snoek, J.; Larochelle, H.; Adams, R. P. Practical Bayesian Optimization of Machine Learning Algorithms. Advances in Neural Information Processing Systems 2012, 25, 2951–2959. https://arxiv.org/abs/1206.2944

[47] Liang, Q.; Gongora, A. E.; Ren, Z.; et al. Benchmarking the Performance of Bayesian Optimization across Multiple Experimental Materials Science Domains. npj Computational Materials 2021, 7, 188. https://doi.org/10.1038/s41524-021-00656-9

6. Appendix

AI Chat PDF Links:

Throughout the research, we ensured ethical use of AI and did not use AI for ghostwriting. A complete AI chat record can be found on this Google Drive:

https://drive.google.com/drive/folders/1vjbmMxIyrG1gq_dguK-B_uqnAvo7Jquz?usp=drive_link

SEM imaging

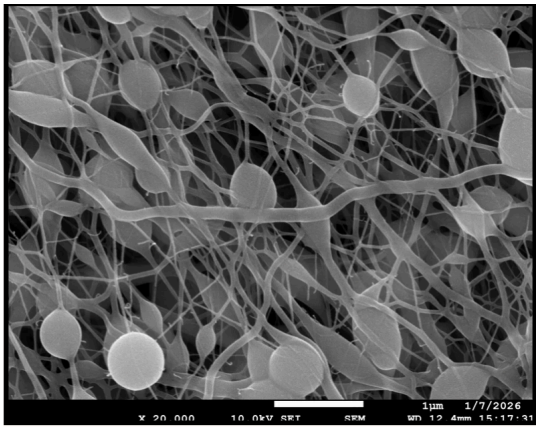


Figure 31. SEM imaging of 14 wt% polymer in the electrospinning dope.

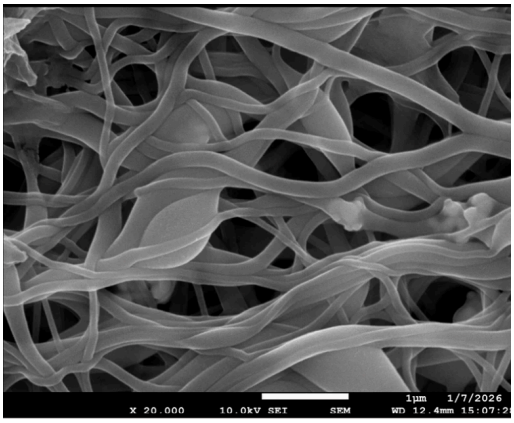


Figure 32. SEM imaging of electrospun matrix (80:20 PVA: PEO ratio)

Table 1: Shade Coefficient Dataset

Shade Coefficient	Dimensions/ mm

	Trapezium base (x_b)	Trapezium top width (x_s)	Vertical spacing (y)	Strand thickness (t)
0.21	10.00	5.00	6.50	1.20
0.30	7.00	4.00	5.50	1.10
0.38	4.67	2.11	5.00	1.05
0.51	2.80	1.50	4.30	1.0

Table 2: Fog water collection for varying shade coefficients for the PLA backbone

Shade coefficient	**Water Collected/ mg**			
	Trial 1	**Trial 2**	**Trial 3**	**Mean ± SD**
0.21	211	21.6	224	217.0 ± 6.6
0.30	231	24.0	226	232.3 ± 7.1
0.38	**256**	**263**	**250**	**256.3 ± 6.5**
0.51	235	238	24.	238.3 ± 3.5

Table 3: Polymer Formulation (polymer wt%) and the qualitative behaviour during electrospinning

Total polymer	**PVA (g)**	**PEO (g)**	**Recorded behaviour**

14 wt%	12.6	1.4	Unstable jet; irregular spinning/beads
16 wt%	14.4	1.6	More stable Taylor cone; some beads
18 wt%	16.2	1.8	Stable Taylor cone; uniform, bead-free fibres
20 wt%	18.0	2.0	Increased viscosity; jet instability/clogging
22 wt%	19.8	2.2	Very viscous; difficult jet formation

Table 4: Polymer Formulation (PVA: PEO ratio) and the qualitative behaviour during electrospinning

PVA: PEO	**PVA in 18 wt% formulation (g)**	**PEO (g)**	**Interpretation**
95:5	17.1	0.9	Lower PEO contribution to jet stabilisation
90:10	16.2	1.8	Selected optimum
85:15	15.3	2.7	Increased instability/bead formation recorded
80:20	14.4	3.6	Softer fibres; composition not retained

Needle to Tip Distance

Table 5: Effect of Needle-to-Tip Distance on Fog Water Collection

Tip-to-collector distance (cm)	**Fog water collection rate/ mg cm^{-2}**

	Trial 1	Trial 2	Trial 3	Mean ± SD
5.0	42	44	45	43.7 ± 1.5
7.5	**50**	**52**	**52**	**51.3 ± 1.2**
10.0	45	47	46	46.0 ± 1.0

Table 6: Increasing TiO_2 loading against Water Collection Rate

TiO_2 loading	Trial 1 (mg cm^{-2} h^{-1})	Trial 2 (mg cm^{-2} h^{-1})	Trial 3 (mg cm^{-2} h^{-1})	Mean ± SD (mg cm^{-2} h^{-1})
0 wt%	53.0	55.4	45.1	51.17 ± 5.39
1 wt%	72.1	75.7	76.2	74.67 ± 2.24
2 wt%	70.3	71.2	71.0	70.83 ± 0.47
3 wt%	69.1	68.7	69.6	69.13 ± 0.45

Table 7: Effect of SiO_2 loading on the maximum force sustained by the PVA/PEO matrix

SiO_2 loading (wt%)	Trial 1, F_{max}	Trial 2, F_{max}	Trial 3, F_{max}	Mean ± SD, F_{max}	Relative to 0 wt%

0	20.76	22.60	23.18	22.18 ± 1.26	1.00
1	30.30	27.52	31.57	29.80 ± 2.07	1.34
2	23.85	26.27	25.61	25.24 ± 1.25	1.14
3	16.79	19.96	20.53	19.09 ± 2.02	0.86

Table 8: Fog water collection rate across the combined TiO_2 and SiO_2 formulation

Formulation	**TiO_2**	**SiO_2**	**Total oxide**	**T1**	**T2**	**T3**	**Mean ± SD**
A	0%	0%	0%	45	53	55	51.0 ± 5.3
B	1%	0%	1%	62	71	73	68.7 ± 5.9
C	1%	1%	2%	64	74	76	71.3 ± 6.4
D	2%	1%	3%	65	76	78	73.0 ± 7.0
E	1%	3%	4%	59	70	71	66.7 ± 6.7

Table 9: Fog Chamber Data for the Raschel mesh, PLA backbone, PVA/PEO matrix, and EcoFog

Collector	**Collection rate (mg cm^{-2} h^{-1})**	**Relative to Raschel**
Raschel mesh	17.7 ± 1.4	1.00

Bare PLA lattice	20.7 ± 1.6	1.17
PVA/PEO Matrix	51.0 ± 5.3	2.88
EcoFog	71.3 ± 6.4	4.03

Table 10: Five-day simultaneous outdoor fog water collection testing for the raschel mesh, PLA backbone, and EcoFog

System	**Accounted water (L m^{-2} day^{-1})**	**Recovered water (L m^{-2} day^{-1})**	**Retained water (L m^{-2} day^{-1})**	**Recovery fraction**
Raschel Mesh	7.4 ± 0.81	4.6 ± 0.51	2.8 ± 0.31	0.622
PLA lattice	7.9 ± 0.82	5.8 ± 0.60	2.1 ± 0.22	0.734
EcoFog	21.4 ± 2.67	17.3 ± 2.10	4.1 ± 0.57	0.808

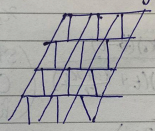
7. Lab Logbook

July 10th 2025 Thursday

Backbone of Material.

↓

Rascral style → Trapezoidal geometry



3D printed PLA.

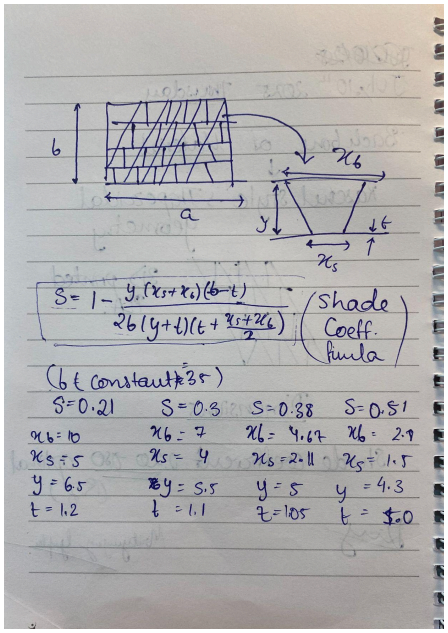
Dimensions

Shade coefficient 0.20 - 0.50 optimal (Ref)

Dey

Mrityunjay Gupta

ACUPHAR MACLEODS



$$S = 1 - \frac{y(x_5+x_6)(b-t)}{2b(y+t)(t+\frac{x_5+x_6}{2})}$$ (Shade Coeff. formula)

(b & constant = 35)

S = 0.21	S = 0.3	S = 0.38	S = 0.51
x_6 = 10	x_6 = 7	x_6 = 4.67	x_6 = 2.9
x_5 = 5	x_5 = 4	x_5 = 2.11	x_5 = 1.5
y = 6.5	y = 5.5	y = 5	y = 4.3
t = 1.2	t = 1.1	t = 1.05	t = 1.0

July 22nd 2025 Tuesday

Water Collection of Varius Shade Coeff.

S	Water Collected/mg T1	T2	T3	Avg.
0.21	211	216	224	217
0.30	231	240	226	232.3
0.38	256	263	250	256.3
0.51	235	239	242	238.3

0.38 shade coefficient achieves highest water collection.

Dey

Mrityunjay Gupta

Aug. 10 2025 Nivedas Materials

PVA & Polymers

PEO

SiO_2 & Nano fillers

TiO_2

~~Matrix~~ Electrospun onto PLA backbone.

90:10 Ratio of PVA to PEO

Varied Concentrations of PVA and PEO Matrix solutions prepared

Dey

Mrityunjay Gupta

OPTIMIZATION @ WT%.

Total wt%	PVA(g)	PEO(g)
14	12.6	1.4
16	14.4	1.6
18	16.2	1.8
20	18.0	2.0
22	19.8	2.2

Solution Preparation

1. Swelling of PVA in water
↓
2. Heat till 90°C until clear solution under continuous stirring
↓
3. Cool to 80°C, add PEO and continue stirring.

~~OPTIMIZATION OF RATIO~~

Aug. 12. 2025 Tuesday

Electrospinning of various matrix. ~~solutions~~

Concentrations tested

14%, 16%, 18%.

14% produced an unstable jet and was similar to 'electrospraying' with irregular whipping. The matrix was irregular and very brittle.

16%, Consistent Taylor Cone, but jet instability remained. Fewer Beads observed however, nanofibers were still non-uniform.

18% - Consistent Taylor Cone formation. Jet formed was stable which led to uniform bead free nanofibers.

Aug. 25. 2025 Monday

Remaining solutions were made

20%, 22%.

20% - Jet was unstable and clogging of needle took place frequently.

22% - ~~No~~ Minimal Taylor Cone formation was seen. Very difficult to form jet.

18 wt% was the optimal concentration.

OPTIMIZATION OF RATIO

Sep. 10. 2025 Wednesday

- TiO_2 incorporated into electrospinning solution. After Ratio of polymers decided

Ratios → PVA:PEO
95:5
85:15
80:20

Optimisation of Polymer ~~Concentration~~ ratio.

Solution 95:5 (18 wt%) for all

PVA(g)	PEO(g)
17.1	0.9

Electrospinning revealed higher jet instability and bead formation.

Sep. 15. 2025 Monday

Solution 85:15

PVA (g)	PEO (g)
15.3	2.7

Electrospinning showcases very stable Taylor Cone fmation, however, the jet fomation was very thin.

Sep. 18. 2025 Thursday

final Solution tested 80:20

PVA (g)	PEO (g)
14.4	3.6

- Soft nanofibers and very easily electrospun. However, very difficult to maintain material attenuation.

Hence, 90:10 was optimal

Sep. 23. 2025 Tuesday

Electrospinning Parameters for optimised Matrix

Optimal voltage → 18kV - 20kV

Needle to Sub. → 18 cm distance

Flow Rate - 0.5 ml/h

Deposition time - 30 minutes

Oct. 1. 2025 Wednesday

Fog Chamber

Ultrasonic Humidifier

Humidity probe

Acrylic Chamber

↙

Acrylic sheets 12X12 inches

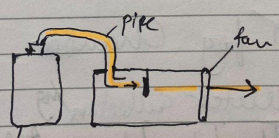
Dimensions

12 x 6 x 6

Sealed with heavy duty glue

Oct. 6. 2025 Monday

- Ultrasonic humidifier arrived
- Testing of Chamber.
- Pipe connected to chamber
- Proper sealing of pipe, so that fog does not escape.
- Chamber was further sealed using glue to prevent leakage of fog.
- Exhaust fan added, simulating wind conditions.



Mrityunjay Gupta

Oct. 9. 2025 Thursday

- Varying conc. of TiO_2 tested inside matrix
  - 0% (Only Matrix)
  - 1%
  - 2%
  - 3%

Solutions proceed
- 0% and 1%

↓

Electrospun onto backbone

↓

Tested in fog Chamber

Conc.	Water Collection (mg)			
	T_1	T_2	T_3	Avg
0%	530	554	451	511.6
1%	721	757	762	746.6

ACUPHAR NUCLEODS

Mrityunjay Gupta

Oct. 14. 2025 Tuesday

Remaining conc. of TiO_2 was tested.

2% & 3%.

Conc.	Water Collection (mg)			Avg
	T_1	T_2	T_3	
2%	~~706~~ 703	712	710	708.3
3%	691	687	696	692.5

Mrityunjay Gupta

Oct. 17. 2025

- 1% Colloidal Silica introduced.

↓

Base matrix

↓

Electrospun onto PLA Backbone

↓

Higher tensile strength as compared to Base matrix with TiO_2 only.

Mrityunjay Gupta

Dec. 2. 2025 Tuesday

- Material sent for Characterisation
FESEM, XRD, Contact Angle, FTIR,
- Results recieved today

Dec. 15. 2025 Monday

Eco Fog

water collection/mg

T1	T2	T3	Avg
1002	1009	1054	1025

$\frac{1025 mg}{10 cm^2}$ = 102.5 mg/cm²

≈ 24 $Lm^{-2} day^{-1}$

102.5 ± 2.82 mg $cm^{-2} h^{-1}$

Rasched Mesh

Water Collected/mg

T1	T2	T3	Avg
378	320	467	383

38.8 ± 7.41 mg $cm^{-2} h^{-1}$

PLA Backbone

Water Collected/mg

T1	T2	T3	Avg
408	431	398	412.3

41.23 ± 1.69 mg $cm^{-2} h^{-1}$

Date

Particle Conc./ mg(TiO2)	Water Collection/mg cm^{-2} T1	T2	T3
0	45.0	53	55
1	62	71	73
2	64	74	76
3	65	76	78
4	59	70.1	71.0

Water Collected / Lm^{-2}

9PM-9AM	9AM-9PM
3.1	1.6

Rasched Mesh

Retained Water - 2.7 Lm^{-2}

Total intercepted - 7.4 Lm^{-2}

Retained Fraction - 36.5%

PLA Skeleton

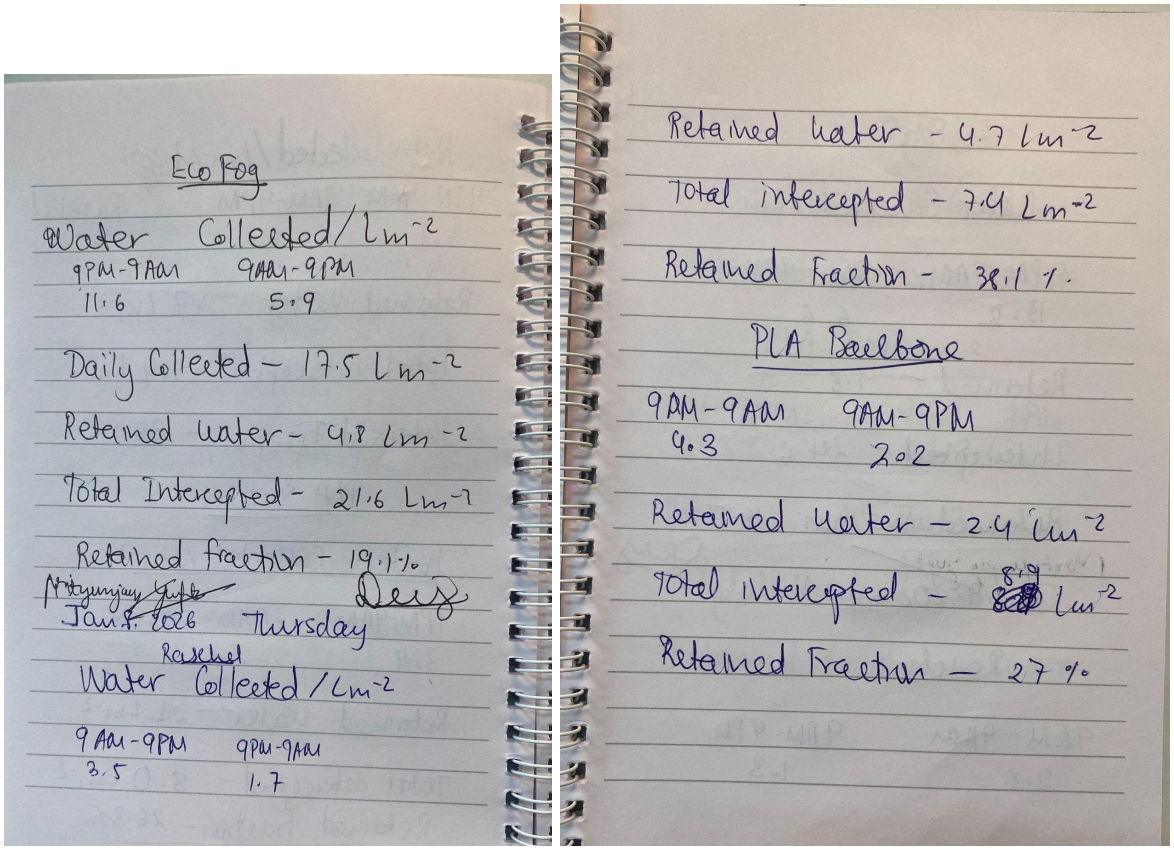
Water Collected / Lm^{-2}

9PM-9AM	9AM-9PM
3.9	2.0

Retained water - 2.1 Lm^{-2}

Total intercepted - 8.0 Lm^{-2}

Retained Fraction - 26.8%



Eco Fog

Water Collected/Lm^{-2}

9PM-9AM	9AM-9PM
11.6	5.9

Daily Collected – 17.5 Lm^{-2}

Retained water – 4.8 Lm^{-2}

Total Intercepted – 21.6 Lm^{-2}

Retained fraction – 19.1%

Priyanjay Gupta Dey

Jan 8, 2026 Thursday

Rescheduled

Water Collected/Lm^{-2}

9AM-9PM	9PM-9AM
3.5	1.7

Retained water – 4.7 Lm^{-2}

Total intercepted – 7.4 Lm^{-2}

Retained Fraction – 38.1 %

PLA Backbone

9PM-9AM	9AM-9PM
4.3	2.2

Retained water – 2.4 Lm^{-2}

Total intercepted – 8.9 Lm^{-2}

Retained Fraction – 27 %

Eco Fog

Water Collected L/m^{-2}

9PM-9AM 9AM-9PM

13.2 6.6

Retained - 4.8

Intercepted - 24.6

Retained Fraction - 19.5 %

Mrityunjay Yadav

Jan, 9, 2026 Friday

Raschel

9PM-9AM 9AM-9PM

2.6 1.3

Retained - 2.4 Lm^{-2}

Intercepted - 6.3 Lm^{-2}

Retained Fraction - 38.1 %

PLA Backbone

9PM-9AM 9AM-9PM

3.3 Lm^{-2} 1.7 Lm^{-2}

Retained - 1.8 Lm^{-2}

Total Intercepted - 6.8

Retained fraction - 26.5 %

Eco Fog

9PM-9AM 9AM-9PM

9.6 Lm^{-2} 5.2 Lm^{-2}

Retained water - 3.1 Lm^{-2}

Intercepted - 18.2 Lm^{-2}

Retained fraction - 18.7 %

Mrityunjay Yadav

Jan 10, 2026 Saturday

Raschel

9PM-9AM 9AM-9PM

3.3 Lm^{-2} 1.6 Lm^{-2}

Retained - 3.0 Lm^{-2}

intercepted - 7.9 Lm^{-2}

Retained fraction - 38.5 %

PLA Backbone

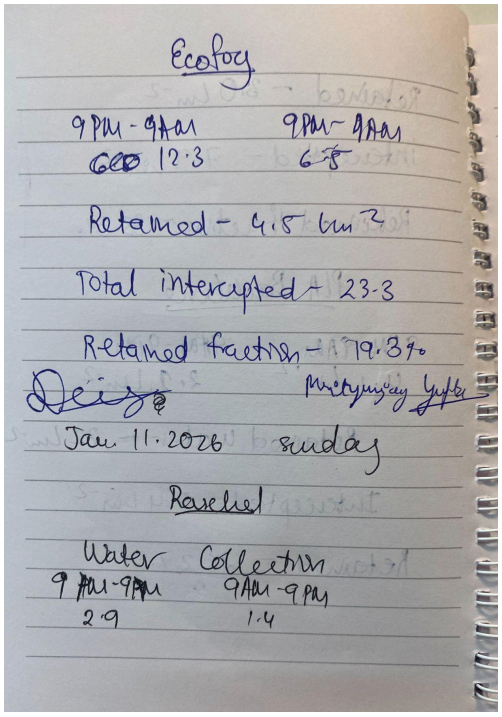
9PM-9AM 9AM-9PM

4.1 Lm^{-2} 2.1 Lm^{-2}

Retained water - 2.2 Lm^{-2}

Intercepted - 9.4 Lm^{-2}

Retained - 26.2 %



Ecofog

9PM - 9AM 9PM - 9AM
12·3 6·8

Retained - 4.5 Lm^{-2}

Total intercepted - 23·3

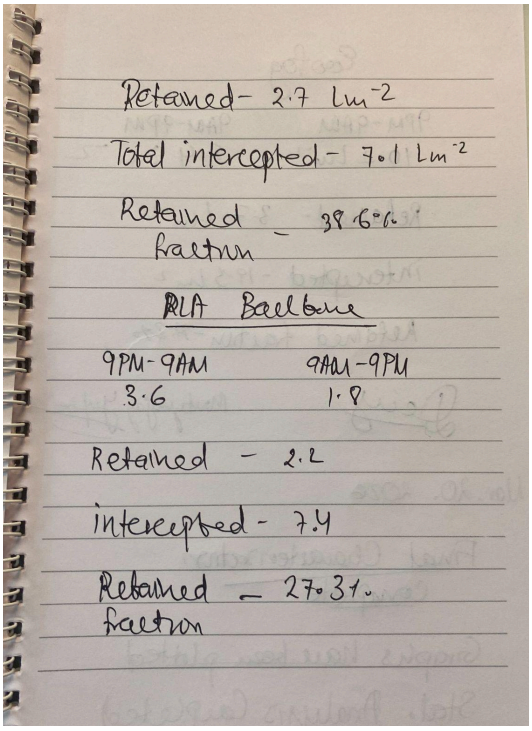
Retained fraction - 19.3%

Mrityunjay Yadav

Jan 11.2026 sunday

Rachel

Water Collection
9PM-9PM 9AM-9PM
2·9 1·4



Retained - 2.7 Lm^{-2}

Total intercepted - 7.1 Lm^{-2}

Retained fraction - 38.6%

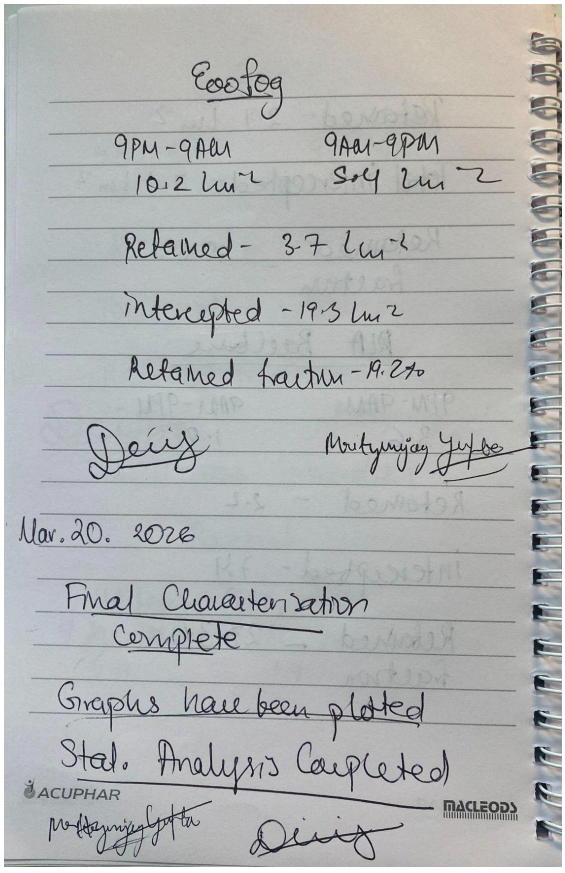
RLA Backbone

9PM-9AM 9AM-9PM
3·6 1·8

Retained - 2.2

intercepted - 7.4

Retained fraction - 27.3%



Ecofog

9PM-9AM 9AM-9PM
10.2 Lm^{-2} 5.4 Lm^{-2}

Retained - 3.7 Lm^{-2}

intercepted - 19.3 Lm^{-2}

Retained fraction - 19.2%

Mrityunjay Yadav

Mar. 20. 2026

Final Characterization complete

Graphs have been plotted

Stat. Analysis Completed

ACUPHAR MACLEODS